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Exercise safety prediction based on physiological conditions

Abstract

Described are techniques, processes, devices, computer-readable media that enable provision of an indication of whether it is safe for a person with diabetes to participate in exercise while using a wearable drug delivery system. A processor may receive or obtain physiological data related to a condition of a wearer of the wearable drug delivery system and by evaluating an exercise model that uses inputs related to the physiological data to make the determination of whether it is safe to exercise and output an exercise safety signal. Modifications to the wearer's medication treatment plan and other actions may be based on an outputted exercise safety signal.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
303013	12/1883	Horton	N/A	N/A
441663	12/1889	Hofbauer	N/A	N/A
955911	12/1909	Saegmuller	N/A	N/A
1441508	12/1922	Marius	N/A	N/A
2797149	12/1956	Skeggs	N/A	N/A
2987214	12/1960	Radack	N/A	N/A
3579805	12/1970	Kast	N/A	N/A
3631847	12/1971	Hobbs	N/A	N/A
3634039	12/1971	Brondy	N/A	N/A
3812843	12/1973	Wootten et al.	N/A	N/A
3841328	12/1973	Jensen	N/A	N/A
3963380	12/1975	Thomas, Jr. et al.	N/A	N/A
4055175	12/1976	Clemens et al.	N/A	N/A
4146029	12/1978	Ellinwood, Jr.	N/A	N/A
4151845	12/1978	Clemens	N/A	N/A
4206401	12/1979	Meyer	N/A	N/A
4245634	12/1980	Albisser et al.	N/A	N/A
4277226	12/1980	Archibald	N/A	N/A
4307713	12/1980	Galkin et al.	N/A	N/A
4368980	12/1982	Aldred et al.	N/A	N/A
4373527	12/1982	Fischell	N/A	N/A
4398542	12/1982	Cunningham et al.	N/A	N/A
4403984	12/1982	Ash et al.	N/A	N/A
4464170	12/1983	Clemens et al.	N/A	N/A
4469481	12/1983	Kobayashi	N/A	N/A
4475901	12/1983	Kraegen et al.	N/A	N/A
4526568	12/1984	Clemens et al.	N/A	N/A
4526569	12/1984	Bernardi	N/A	N/A
4529401	12/1984	Leslie et al.	N/A	N/A
4559033	12/1984	Stephen et al.	N/A	N/A
4559037	12/1984	Franetzki et al.	N/A	N/A
4560979	12/1984	Roskopf	N/A	N/A
4573968	12/1985	Parker	N/A	N/A
4587850	12/1985	Moser	N/A	N/A
4624661	12/1985	Arimond	N/A	N/A
4633878	12/1986	Bombardieri	N/A	N/A
4657529	12/1986	Prince et al.	N/A	N/A
4685903	12/1986	Cable et al.	N/A	N/A
4731726	12/1987	Allen, III	N/A	N/A
4743243	12/1987	Vaillancourt	N/A	N/A
4755173	12/1987	Konopka et al.	N/A	N/A
4781688	12/1987	Thoma et al.	N/A	N/A
4781693	12/1987	Martinez et al.	N/A	N/A
4801957	12/1988	Vandemoere	N/A	N/A
4808161	12/1988	Kamen	N/A	N/A
4833088	12/1988	Desimone et al.	N/A	N/A
4836752	12/1988	Burkett	N/A	N/A
4850954	12/1988	Charvin	N/A	N/A
4854170	12/1988	Brimhall et al.	N/A	N/A
4882600	12/1988	Van de Moere	N/A	N/A
4886499	12/1988	Cirelli et al.	N/A	N/A
4900292	12/1989	Berry et al.	N/A	N/A
4919596	12/1989	Slate et al.	N/A	N/A
4925444	12/1989	Orkin et al.	N/A	N/A
4940527	12/1989	Kazlauskas et al.	N/A	N/A
4961055	12/1989	Habib et al.	N/A	N/A
4973998	12/1989	Gates	N/A	N/A
4975581	12/1989	Robinson et al.	N/A	N/A

4976720	12/1989	Machold et al.	N/A	N/A
4981140	12/1990	Wyatt	N/A	N/A
4994047	12/1990	Walker et al.	N/A	N/A
5007286	12/1990	Malcolm et al.	N/A	N/A
5045871	12/1990	Reinholdson	N/A	N/A
5097834	12/1991	Skrabal	N/A	N/A
5102406	12/1991	Arnold	N/A	N/A
5109850	12/1991	Blanco et al.	N/A	N/A
5125415	12/1991	Bell	N/A	N/A
5134079	12/1991	Cusack et al.	N/A	N/A
5153827	12/1991	Coutre et al.	N/A	N/A
5165406	12/1991	Wong	N/A	N/A
5176662	12/1992	Bartholomew et al.	N/A	N/A
5178609	12/1992	Ishikawa	N/A	N/A
5207642	12/1992	Orkin et al.	N/A	N/A
5232439	12/1992	Campbell et al.	N/A	N/A
5232668	12/1992	Grant et al.	N/A	N/A
5237993	12/1992	Skrabal	N/A	N/A
5239326	12/1992	Takai	N/A	N/A
5244459	12/1992	Hill	N/A	N/A
5244463	12/1992	Cordner, Jr. et al.	N/A	N/A
5257980	12/1992	Van Antwerp et al.	N/A	N/A
5273517	12/1992	Barone et al.	N/A	N/A
5281808	12/1993	Kunkel	N/A	N/A
5299571	12/1993	Mastrototaro	N/A	N/A
5308982	12/1993	Ivaldi et al.	N/A	N/A
5342298	12/1993	Michaels et al.	N/A	N/A
5377674	12/1994	Kuestner	N/A	N/A
5380665	12/1994	Cusack et al.	N/A	N/A
5385539	12/1994	Maynard	N/A	N/A
5389078	12/1994	Zalesky	N/A	N/A
5411889	12/1994	Hoots et al.	N/A	N/A
5421812	12/1994	Langley et al.	N/A	N/A
5452033	12/1994	Balling et al.	N/A	N/A
5468727	12/1994	Phillips et al.	N/A	N/A
5505709	12/1995	Funderburk et al.	N/A	N/A
5505828	12/1995	Wong et al.	N/A	N/A
5507288	12/1995	Bocker et al.	N/A	N/A
5533389	12/1995	Kamen et al.	N/A	N/A
5558640	12/1995	Pfeiler et al.	N/A	N/A
5563584	12/1995	Rader et al.	N/A	N/A
5569186	12/1995	Lord et al.	N/A	N/A
5576781	12/1995	Deleeuw	N/A	N/A
5584813	12/1995	Livingston et al.	N/A	N/A
5585733	12/1995	Paglione	N/A	N/A
5586553	12/1995	Halili et al.	N/A	N/A
5609572	12/1996	Lang	N/A	N/A
5647853	12/1996	Feldmann et al.	N/A	N/A
5660163	12/1996	Schulman et al.	N/A	N/A
5665065	12/1996	Colman et al.	N/A	N/A
5678539	12/1996	Schubert et al.	N/A	N/A
5685844	12/1996	Marttila	N/A	N/A
5685859	12/1996	Kornerup	N/A	N/A
5693018	12/1996	Kriesel et al.	N/A	N/A
5697899	12/1996	Hillman et al.	N/A	N/A
5700695	12/1996	Yassinzadeh et al.	N/A	N/A
5703364	12/1996	Rosenthal	N/A	N/A
5714123	12/1997	Sohrab	N/A	N/A
5716343	12/1997	Kriesel et al.	N/A	N/A
5722397	12/1997	Eppstein	N/A	N/A
5726404	12/1997	Brody	N/A	N/A
5726751	12/1997	Altendorf et al.	N/A	N/A
5741228	12/1997	Lambrecht et al.	N/A	N/A

5746217	12/1997	Erickson et al.	N/A	N/A
5755682	12/1997	Knudson et al.	N/A	N/A
5758643	12/1997	Wong et al.	N/A	N/A
5785681	12/1997	Indravudh et al.	N/A	N/A
5800405	12/1997	McPhee	N/A	N/A
5800420	12/1997	Gross et al.	N/A	N/A
5801057	12/1997	Smart et al.	N/A	N/A
5804048	12/1997	Wong et al.	N/A	N/A
5817007	12/1997	Fodgaard et al.	N/A	N/A
5820622	12/1997	Gross et al.	N/A	N/A
5823951	12/1997	Messerschmidt	N/A	N/A
5830999	12/1997	Dunn	N/A	N/A
5840020	12/1997	Heinonen et al.	N/A	N/A
5848991	12/1997	Gross et al.	N/A	N/A
5851197	12/1997	Marano et al.	N/A	N/A
5858005	12/1998	Kriesel	N/A	N/A
5865806	12/1998	Howell	N/A	N/A
5867688	12/1998	Simmon et al.	N/A	N/A
5871470	12/1998	McWha	N/A	N/A
5879310	12/1998	Sopp et al.	N/A	N/A
5899882	12/1998	Waksman et al.	N/A	N/A
5902253	12/1998	Pfeiffer et al.	N/A	N/A
5931814	12/1998	Alex et al.	N/A	N/A
5932175	12/1998	Knute et al.	N/A	N/A
5935099	12/1998	Peterson et al.	N/A	N/A
5947911	12/1998	Wong et al.	N/A	N/A
5971941	12/1998	Simons et al.	N/A	N/A
5993423	12/1998	Choi	N/A	N/A
5995236	12/1998	Roth et al.	N/A	N/A
5997501	12/1998	Gross et al.	N/A	N/A
6017318	12/1999	Gauthier et al.	N/A	N/A
6024539	12/1999	Blomquist	N/A	N/A
6032059	12/1999	Henning et al.	N/A	N/A
6036924	12/1999	Simons et al.	N/A	N/A
6040578	12/1999	Malin et al.	N/A	N/A
6049727	12/1999	Crothall	N/A	N/A
6050978	12/1999	Orr et al.	N/A	N/A
6058934	12/1999	Sullivan	N/A	N/A
6066103	12/1999	Duchon et al.	N/A	N/A
6071292	12/1999	Makower et al.	N/A	N/A
6072180	12/1999	Kramer et al.	N/A	N/A
6077055	12/1999	Vilks	N/A	N/A
6090092	12/1999	Fowles et al.	N/A	N/A
6101406	12/1999	Hacker et al.	N/A	N/A
6102872	12/1999	Doneen et al.	N/A	N/A
6115673	12/1999	Malin et al.	N/A	N/A
6123827	12/1999	Wong et al.	N/A	N/A
6124134	12/1999	Stark	N/A	N/A
6126637	12/1999	Kriesel et al.	N/A	N/A
6128519	12/1999	Say	N/A	N/A
6142181	12/1999	Schumacher	N/A	N/A
6142939	12/1999	Eppstein et al.	N/A	N/A
6143164	12/1999	Heller et al.	N/A	N/A
6157041	12/1999	Thomas et al.	N/A	N/A
6161028	12/1999	Braig et al.	N/A	N/A
6162639	12/1999	Douglas	N/A	N/A
6171264	12/2000	Bader	N/A	N/A
6196046	12/2000	Braig et al.	N/A	N/A
6200287	12/2000	Keller et al.	N/A	N/A
6200293	12/2000	Kriesel et al.	N/A	N/A
6200338	12/2000	Solomon et al.	N/A	N/A
6214629	12/2000	Freitag et al.	N/A	N/A
6226082	12/2000	Roe	N/A	N/A

6244776	12/2000	Wiley	N/A	N/A
6261065	12/2000	Nayak et al.	N/A	N/A
6262798	12/2000	Shepherd et al.	N/A	N/A
6270455	12/2000	Brown	N/A	N/A
6271045	12/2000	Douglas et al.	N/A	N/A
6280381	12/2000	Malin et al.	N/A	N/A
6285448	12/2000	Kuenstner	N/A	N/A
6309370	12/2000	Haim et al.	N/A	N/A
6312888	12/2000	Wong et al.	N/A	N/A
6334851	12/2001	Hayes et al.	N/A	N/A
6375627	12/2001	Mauze et al.	N/A	N/A
6379301	12/2001	Worthington et al.	N/A	N/A
6381029	12/2001	Tipirneni	N/A	N/A
6402689	12/2001	Scarantino et al.	N/A	N/A
6418332	12/2001	Mastrototaro et al.	N/A	N/A
6470279	12/2001	Samsoondar	N/A	N/A
6475196	12/2001	Vachon	N/A	N/A
6477901	12/2001	Tadigadapa et al.	N/A	N/A
6484044	12/2001	Lilienfeld-Toal	N/A	N/A
6491656	12/2001	Morris	N/A	N/A
6512937	12/2002	Blank et al.	N/A	N/A
6514460	12/2002	Fendrock	N/A	N/A
6525509	12/2002	Petersson et al.	N/A	N/A
6528809	12/2002	Thomas et al.	N/A	N/A
6540672	12/2002	Simonsen et al.	N/A	N/A
6544212	12/2002	Galley et al.	N/A	N/A
6546268	12/2002	Ishikawa et al.	N/A	N/A
6546269	12/2002	Kurnik	N/A	N/A
6553841	12/2002	Blouch	N/A	N/A
6554798	12/2002	Mann et al.	N/A	N/A
6556850	12/2002	Braig et al.	N/A	N/A
6558351	12/2002	Steil et al.	N/A	N/A
6560471	12/2002	Heller et al.	N/A	N/A
6561978	12/2002	Conn et al.	N/A	N/A
6562001	12/2002	Lebel et al.	N/A	N/A
6562014	12/2002	Lin et al.	N/A	N/A
6569125	12/2002	Jepson et al.	N/A	N/A
6572542	12/2002	Houben et al.	N/A	N/A
6572545	12/2002	Knobbe et al.	N/A	N/A
6574490	12/2002	Abbink et al.	N/A	N/A
6575905	12/2002	Knobbe et al.	N/A	N/A
6580934	12/2002	Braig et al.	N/A	N/A
6618603	12/2002	Varalli et al.	N/A	N/A
6633772	12/2002	Ford et al.	N/A	N/A
6645142	12/2002	Braig et al.	N/A	N/A
6653091	12/2002	Dunn et al.	N/A	N/A
6662030	12/2002	Khalil et al.	N/A	N/A
6669663	12/2002	Thompson	N/A	N/A
6678542	12/2003	Braig et al.	N/A	N/A
6685452	12/2003	Christiansen et al.	N/A	N/A
6699221	12/2003	Vaillancourt	N/A	N/A
6718189	12/2003	Rohrscheib et al.	N/A	N/A
6721582	12/2003	Trepagnier et al.	N/A	N/A
6728560	12/2003	Kollias et al.	N/A	N/A
6740059	12/2003	Flaherty	N/A	N/A
6740072	12/2003	Starkweather et al.	N/A	N/A
6751490	12/2003	Esenaliev et al.	N/A	N/A
6758835	12/2003	Close et al.	N/A	N/A
6768319	12/2003	Wang	N/A	N/A
6768425	12/2003	Flaherty et al.	N/A	N/A
6780156	12/2003	Haueter et al.	N/A	N/A
6810290	12/2003	Lebel et al.	N/A	N/A
6830558	12/2003	Flaherty et al.	N/A	N/A

6837858	12/2004	Cunningham et al.	N/A	N/A
6837988	12/2004	Leong et al.	N/A	N/A
6846288	12/2004	Nagar et al.	N/A	N/A
6862534	12/2004	Sterling et al.	N/A	N/A
6865408	12/2004	Abbink et al.	N/A	N/A
6890291	12/2004	Robinson et al.	N/A	N/A
6936029	12/2004	Mann et al.	N/A	N/A
6949081	12/2004	Chance	N/A	N/A
6958809	12/2004	Sterling et al.	N/A	N/A
6989891	12/2005	Braig et al.	N/A	N/A
6990366	12/2005	Say et al.	N/A	N/A
7008404	12/2005	Nakajima	N/A	N/A
7009180	12/2005	Sterling et al.	N/A	N/A
7016713	12/2005	Gardner et al.	N/A	N/A
7018360	12/2005	Flaherty et al.	N/A	N/A
7025743	12/2005	Mann et al.	N/A	N/A
7025744	12/2005	Utterberg et al.	N/A	N/A
7027848	12/2005	Robinson et al.	N/A	N/A
7043288	12/2005	Davis, III et al.	N/A	N/A
7060059	12/2005	Keith et al.	N/A	N/A
7061593	12/2005	Braig et al.	N/A	N/A
7096124	12/2005	Sterling et al.	N/A	N/A
7115205	12/2005	Robinson et al.	N/A	N/A
7128727	12/2005	Flaherty et al.	N/A	N/A
7137964	12/2005	Flaherty	N/A	N/A
7139593	12/2005	Kavak et al.	N/A	N/A
7139598	12/2005	Hull et al.	N/A	N/A
7144384	12/2005	Gorman et al.	N/A	N/A
7171252	12/2006	Scarantino et al.	N/A	N/A
7182726	12/2006	Williams et al.	N/A	N/A
7190988	12/2006	Say et al.	N/A	N/A
7204823	12/2006	Estes et al.	N/A	N/A
7248912	12/2006	Gough et al.	N/A	N/A
7267665	12/2006	Steil et al.	N/A	N/A
7271912	12/2006	Sterling et al.	N/A	N/A
7278983	12/2006	Ireland et al.	N/A	N/A
7291107	12/2006	Hellwig et al.	N/A	N/A
7291497	12/2006	Holmes et al.	N/A	N/A
7303073	12/2006	Raynal-Olive et al.	N/A	N/A
7303549	12/2006	Flaherty et al.	N/A	N/A
7303622	12/2006	Loch et al.	N/A	N/A
7303922	12/2006	Jeng et al.	N/A	N/A
7354420	12/2007	Steil et al.	N/A	N/A
7388202	12/2007	Sterling et al.	N/A	N/A
7402153	12/2007	Steil et al.	N/A	N/A
7404796	12/2007	Ginsberg	N/A	N/A
7429255	12/2007	Thompson	N/A	N/A
7460130	12/2007	Salganicoff	N/A	N/A
7481787	12/2008	Gable et al.	N/A	N/A
7491187	12/2008	Van Den Berghe et al.	N/A	N/A
7500949	12/2008	Gottlieb et al.	N/A	N/A
7509156	12/2008	Flanders	N/A	N/A
7547281	12/2008	Hayes et al.	N/A	N/A
7569030	12/2008	Ebel et al.	N/A	N/A
7608042	12/2008	Goldberger et al.	N/A	N/A
7651845	12/2009	Doyle, III et al.	N/A	N/A
7680529	12/2009	Kroll	N/A	N/A
7731900	12/2009	Haar et al.	N/A	N/A
7734323	12/2009	Blomquist et al.	N/A	N/A
7766829	12/2009	Sloan et al.	N/A	N/A
7785258	12/2009	Braig et al.	N/A	N/A
7806854	12/2009	Damiano et al.	N/A	N/A
7806886	12/2009	Kanderian, Jr. et al.	N/A	N/A

7842241	12/2009	Arbogast et al.	N/A	N/A
7846385	12/2009	Arbogast et al.	N/A	N/A
7846386	12/2009	Arbogast et al.	N/A	N/A
7846387	12/2009	Arbogast et al.	N/A	N/A
7846388	12/2009	Arbogast et al.	N/A	N/A
7867446	12/2010	Arbogast et al.	N/A	N/A
7897107	12/2010	Arbogast et al.	N/A	N/A
7914742	12/2010	Arbogast et al.	N/A	N/A
7918825	12/2010	OConnor et al.	N/A	N/A
7946985	12/2010	Mastrototaro et al.	N/A	N/A
7972296	12/2010	Braig et al.	N/A	N/A
8003052	12/2010	Sacherer	N/A	N/A
8056719	12/2010	Porret et al.	N/A	N/A
8080205	12/2010	Arbogast et al.	N/A	N/A
8105282	12/2011	Susi et al.	N/A	N/A
8221345	12/2011	Blomquist	N/A	N/A
8251907	12/2011	Sterling et al.	N/A	N/A
8285487	12/2011	Bergstrom et al.	N/A	N/A
8431408	12/2012	Lewis et al.	N/A	N/A
8449524	12/2012	Braig et al.	N/A	N/A
8452359	12/2012	Rebec et al.	N/A	N/A
8454557	12/2012	Qi et al.	N/A	N/A
8454576	12/2012	Mastrototaro et al.	N/A	N/A
8461561	12/2012	Freeman et al.	N/A	N/A
8465977	12/2012	Joseph et al.	N/A	N/A
8467980	12/2012	Campbell et al.	N/A	N/A
8478557	12/2012	Hayter et al.	N/A	N/A
8547239	12/2012	Peatfield et al.	N/A	N/A
8597274	12/2012	Sloan et al.	N/A	N/A
8622988	12/2013	Hayter	N/A	N/A
8701264	12/2013	Martinson	N/A	N/A
8727117	12/2013	Maasarani	N/A	N/A
8765482	12/2013	Joseph et al.	N/A	N/A
8810394	12/2013	Kalpin	N/A	N/A
8894262	12/2013	Celentano et al.	N/A	N/A
9005166	12/2014	Uber, III et al.	N/A	N/A
9061097	12/2014	Holt et al.	N/A	N/A
9171343	12/2014	Fischell et al.	N/A	N/A
9233204	12/2015	Booth et al.	N/A	N/A
9248229	12/2015	Devouassoux et al.	N/A	N/A
9265877	12/2015	Mcarthur et al.	N/A	N/A
9427710	12/2015	Jansen	N/A	N/A
9486571	12/2015	Rosinko	N/A	N/A
9572926	12/2016	Cabiri	N/A	N/A
9579456	12/2016	Budiman et al.	N/A	N/A
9598195	12/2016	Deutschle et al.	N/A	N/A
9743224	12/2016	San Vicente et al.	N/A	N/A
9814832	12/2016	Agard et al.	N/A	N/A
9862519	12/2017	Deutschle et al.	N/A	N/A
9907515	12/2017	Doyle, III et al.	N/A	N/A
9980140	12/2017	Spencer et al.	N/A	N/A
9984773	12/2017	Gondhalekar et al.	N/A	N/A
10046114	12/2017	Biederman et al.	N/A	N/A
RE47100	12/2017	Smith et al.	N/A	N/A
10086131	12/2017	Okihara	N/A	N/A
10248839	12/2018	Levy et al.	N/A	N/A
10335464	12/2018	Michelich et al.	N/A	N/A
10342926	12/2018	Nazzaro et al.	N/A	N/A
10441717	12/2018	Schmid et al.	N/A	N/A
10583250	12/2019	Mazlish et al.	N/A	N/A
10661012	12/2019	Nazzaro et al.	N/A	N/A
10737024	12/2019	Schmid	N/A	N/A
10987468	12/2020	Mazlish et al.	N/A	N/A

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11260169	12/2021	Estes	N/A	N/A
2001/0021803	12/2000	Blank et al.	N/A	N/A
2001/0034023	12/2000	Stanton, Jr. et al.	N/A	N/A
2001/0034502	12/2000	Moberg et al.	N/A	N/A
2001/0051377	12/2000	Hammer et al.	N/A	N/A
2001/0053895	12/2000	Vaillancourt	N/A	N/A
2002/0010401	12/2001	Bushmakina et al.	N/A	N/A
2002/0010423	12/2001	Gross et al.	N/A	N/A
2002/0016568	12/2001	Lebel et al.	N/A	N/A
2002/0032374	12/2001	Holker et al.	N/A	N/A
2002/0040208	12/2001	Flaherty et al.	N/A	N/A
2002/0123740	12/2001	Flaherty et al.	N/A	N/A
2002/0128543	12/2001	Leonhardt	N/A	N/A
2002/0147423	12/2001	Burbank et al.	N/A	N/A
2002/0155425	12/2001	Han et al.	N/A	N/A
2002/0161288	12/2001	Shin et al.	N/A	N/A
2002/0161307	12/2001	Yu et al.	N/A	N/A
2003/0023148	12/2002	Lorenz et al.	N/A	N/A
2003/0050621	12/2002	Lebel et al.	N/A	N/A
2003/0060692	12/2002	Ruchti et al.	N/A	N/A
2003/0073952	12/2002	Flaherty et al.	N/A	N/A
2003/0086074	12/2002	Braig et al.	N/A	N/A
2003/0086075	12/2002	Braig et al.	N/A	N/A
2003/0090649	12/2002	Sterling et al.	N/A	N/A
2003/0100040	12/2002	Bonnecaze et al.	N/A	N/A
2003/0130616	12/2002	Steil et al.	N/A	N/A
2003/0135388	12/2002	Martucci et al.	N/A	N/A
2003/0144582	12/2002	Cohen et al.	N/A	N/A
2003/0163097	12/2002	Fleury et al.	N/A	N/A
2003/0195404	12/2002	Knobbe et al.	N/A	N/A
2003/0208113	12/2002	Mault et al.	N/A	N/A
2003/0208154	12/2002	Close et al.	N/A	N/A
2003/0212379	12/2002	Bylund et al.	N/A	N/A
2003/0216627	12/2002	Lorenz et al.	N/A	N/A
2003/0220605	12/2002	Bowman, Jr. et al.	N/A	N/A
2004/0010207	12/2003	Flaherty et al.	N/A	N/A
2004/0010507	12/2003	Bellew	N/A	N/A
2004/0034295	12/2003	Salganicoff	N/A	N/A
2004/0045879	12/2003	Shults et al.	N/A	N/A
2004/0051368	12/2003	Caputo et al.	N/A	N/A
2004/0064259	12/2003	Haaland et al.	N/A	N/A
2004/0085215	12/2003	Moberg et al.	N/A	N/A
2004/0097796	12/2003	Berman et al.	N/A	N/A
2004/0116847	12/2003	Wall	N/A	N/A
2004/0122353	12/2003	Shahmirian et al.	N/A	N/A
2004/0133166	12/2003	Moberg et al.	N/A	N/A
2004/0147034	12/2003	Gore et al.	N/A	N/A
2004/0171983	12/2003	Sparks et al.	N/A	N/A
2004/0203357	12/2003	Nassimi	N/A	N/A
2004/0204868	12/2003	Maynard et al.	N/A	N/A
2004/0215492	12/2003	Choi	N/A	N/A
2004/0220517	12/2003	Starkweather et al.	N/A	N/A
2004/0241736	12/2003	Hendee et al.	N/A	N/A
2004/0249308	12/2003	Forssell	N/A	N/A
2005/0003470	12/2004	Nelson et al.	N/A	N/A
2005/0009126	12/2004	Andrews et al.	N/A	N/A
2005/0020980	12/2004	Inoue et al.	N/A	N/A
2005/0022274	12/2004	Campbell et al.	N/A	N/A
2005/0033148	12/2004	Haueter et al.	N/A	N/A
2005/0049179	12/2004	Davidson et al.	N/A	N/A
2005/0055242	12/2004	Bello et al.	N/A	N/A
2005/0065464	12/2004	Talbot et al.	N/A	N/A

2005/0065465	12/2004	Lebel et al.	N/A	N/A
2005/0075624	12/2004	Miesel	N/A	N/A
2005/0105095	12/2004	Pesach et al.	N/A	N/A
2005/0125162	12/2004	Hajizadeh et al.	N/A	N/A
2005/0137573	12/2004	McLaughlin	N/A	N/A
2005/0171503	12/2004	Van Den Berghe et al.	N/A	N/A
2005/0182306	12/2004	Sloan	N/A	N/A
2005/0192494	12/2004	Ginsberg	N/A	N/A
2005/0192557	12/2004	Brauker et al.	N/A	N/A
2005/0197621	12/2004	Poulsen et al.	N/A	N/A
2005/0201897	12/2004	Zimmer et al.	N/A	N/A
2005/0203360	12/2004	Brauker et al.	N/A	N/A
2005/0203461	12/2004	Flaherty et al.	N/A	N/A
2005/0232815	12/2004	Ruhl et al.	N/A	N/A
2005/0238507	12/2004	Dilanni et al.	N/A	N/A
2005/0261660	12/2004	Choi	N/A	N/A
2005/0272640	12/2004	Doyle, III et al.	N/A	N/A
2005/0277164	12/2004	Drucker et al.	N/A	N/A
2005/0277912	12/2004	John	N/A	N/A
2006/0009727	12/2005	OMahony et al.	N/A	N/A
2006/0079809	12/2005	Goldberger et al.	N/A	N/A
2006/0086909	12/2005	Schaber	N/A	N/A
2006/0092569	12/2005	Che et al.	N/A	N/A
2006/0100494	12/2005	Kroll	N/A	N/A
2006/0134323	12/2005	OBrien	N/A	N/A
2006/0167350	12/2005	Monfre et al.	N/A	N/A
2006/0173406	12/2005	Hayes et al.	N/A	N/A
2006/0189925	12/2005	Gable et al.	N/A	N/A
2006/0189926	12/2005	Hall et al.	N/A	N/A
2006/0197015	12/2005	Sterling et al.	N/A	N/A
2006/0200070	12/2005	Callicoat et al.	N/A	N/A
2006/0204535	12/2005	Johnson	N/A	N/A
2006/0229531	12/2005	Goldberger et al.	N/A	N/A
2006/0253085	12/2005	Geismar et al.	N/A	N/A
2006/0264895	12/2005	Flanders	N/A	N/A
2006/0264926	12/2005	Kochamba	N/A	N/A
2006/0270983	12/2005	Lord et al.	N/A	N/A
2006/0276771	12/2005	Galley et al.	N/A	N/A
2006/0282290	12/2005	Flaherty et al.	N/A	N/A
2007/0016127	12/2006	Staib et al.	N/A	N/A
2007/0027370	12/2006	Brauker et al.	N/A	N/A
2007/0060796	12/2006	Kim	N/A	N/A
2007/0060869	12/2006	Tolle et al.	N/A	N/A
2007/0060872	12/2006	Hall et al.	N/A	N/A
2007/0078784	12/2006	Donovan et al.	N/A	N/A
2007/0083160	12/2006	Hall et al.	N/A	N/A
2007/0106135	12/2006	Sloan et al.	N/A	N/A
2007/0116601	12/2006	Patton	N/A	N/A
2007/0118405	12/2006	Campbell et al.	N/A	N/A
2007/0129690	12/2006	Rosenblatt et al.	N/A	N/A
2007/0142720	12/2006	Ridder et al.	N/A	N/A
2007/0173761	12/2006	Kanderian et al.	N/A	N/A
2007/0173974	12/2006	Lin et al.	N/A	N/A
2007/0179352	12/2006	Randlov et al.	N/A	N/A
2007/0179885	12/2006	Bird et al.	N/A	N/A
2007/0191716	12/2006	Goldberger et al.	N/A	N/A
2007/0191770	12/2006	Moberg et al.	N/A	N/A
2007/0197163	12/2006	Robertson	N/A	N/A
2007/0225675	12/2006	Robinson et al.	N/A	N/A
2007/0233051	12/2006	Hohl et al.	N/A	N/A
2007/0244381	12/2006	Robinson et al.	N/A	N/A
2007/0249007	12/2006	Rosero	N/A	N/A
2007/0264707	12/2006	Liederman et al.	N/A	N/A

2007/0282269	12/2006	Carter et al.	N/A	N/A
2007/0287985	12/2006	Estes et al.	N/A	N/A
2007/0293843	12/2006	Ireland et al.	N/A	N/A
2008/0004515	12/2007	Jennewine	N/A	N/A
2008/0027371	12/2007	Higuchi et al.	N/A	N/A
2008/0033272	12/2007	Gough et al.	N/A	N/A
2008/0051764	12/2007	Dent et al.	N/A	N/A
2008/0058625	12/2007	McGarraugh et al.	N/A	N/A
2008/0065050	12/2007	Sparks et al.	N/A	N/A
2008/0071157	12/2007	McGarraugh et al.	N/A	N/A
2008/0071158	12/2007	McGarraugh et al.	N/A	N/A
2008/0077081	12/2007	Mounce et al.	N/A	N/A
2008/0078400	12/2007	Martens et al.	N/A	N/A
2008/0097289	12/2007	Steil et al.	N/A	N/A
2008/0132880	12/2007	Buchman	N/A	N/A
2008/0161664	12/2007	Mastrototaro et al.	N/A	N/A
2008/0172026	12/2007	Blomquist	N/A	N/A
2008/0173073	12/2007	Downie et al.	N/A	N/A
2008/0177165	12/2007	Blomquist et al.	N/A	N/A
2008/0188796	12/2007	Steil et al.	N/A	N/A
2008/0200838	12/2007	Goldberger et al.	N/A	N/A
2008/0206067	12/2007	De Corral et al.	N/A	N/A
2008/0208113	12/2007	Damiano et al.	N/A	N/A
2008/0214919	12/2007	Harmon et al.	N/A	N/A
2008/0228056	12/2007	Blomquist et al.	N/A	N/A
2008/0249386	12/2007	Besterman et al.	N/A	N/A
2008/0255438	12/2007	Saidara et al.	N/A	N/A
2008/0269585	12/2007	Ginsberg	N/A	N/A
2008/0269714	12/2007	Mastrototaro et al.	N/A	N/A
2008/0269723	12/2007	Mastrototaro et al.	N/A	N/A
2008/0281290	12/2007	Yodfat et al.	N/A	N/A
2008/0287906	12/2007	Burkholz et al.	N/A	N/A
2009/0006061	12/2008	Thukral et al.	N/A	N/A
2009/0018406	12/2008	Yodfat et al.	N/A	N/A
2009/0030398	12/2008	Yodfat et al.	N/A	N/A
2009/0036753	12/2008	King	N/A	N/A
2009/0043240	12/2008	Robinson et al.	N/A	N/A
2009/0048556	12/2008	Durand	N/A	N/A
2009/0054753	12/2008	Robinson et al.	N/A	N/A
2009/0062767	12/2008	Van Antwerp et al.	N/A	N/A
2009/0069743	12/2008	Krishnamoorthy et al.	N/A	N/A
2009/0069745	12/2008	Estes et al.	N/A	N/A
2009/0069787	12/2008	Estes et al.	N/A	N/A
2009/0099521	12/2008	Gravesen et al.	N/A	N/A
2009/0105573	12/2008	Malecha	N/A	N/A
2009/0112769	12/2008	Dicks et al.	N/A	N/A
2009/0131861	12/2008	Braig et al.	N/A	N/A
2009/0156922	12/2008	Goldberger et al.	N/A	N/A
2009/0156924	12/2008	Shariati et al.	N/A	N/A
2009/0163781	12/2008	Say et al.	N/A	N/A
2009/0177142	12/2008	Blomquist et al.	N/A	N/A
2009/0198350	12/2008	Thiele	N/A	N/A
2009/0204078	12/2008	Mitchell et al.	N/A	N/A
2009/0221890	12/2008	Saffer et al.	N/A	N/A
2009/0228214	12/2008	Say et al.	N/A	N/A
2009/0254041	12/2008	Krag et al.	N/A	N/A
2009/0282947	12/2008	Powell	N/A	N/A
2009/0318791	12/2008	Kaastруп	N/A	N/A
2009/0326343	12/2008	Gable et al.	N/A	N/A
2010/0057042	12/2009	Hayter	N/A	N/A
2010/0076275	12/2009	Chu et al.	N/A	N/A
2010/0094251	12/2009	Estes	N/A	N/A
2010/0114026	12/2009	Karratt et al.	N/A	N/A

2010/0121170	12/2009	Rule	N/A	N/A
2010/0137784	12/2009	Cefai et al.	N/A	N/A
2010/0145272	12/2009	Cefai et al.	N/A	N/A
2010/0152658	12/2009	Hanson et al.	N/A	N/A
2010/0168683	12/2009	Cabiri	N/A	N/A
2010/0174228	12/2009	Buckingham et al.	N/A	N/A
2010/0185175	12/2009	Kamen et al.	N/A	N/A
2010/0211003	12/2009	Sundar et al.	N/A	N/A
2010/0228110	12/2009	Tsoukalis	N/A	N/A
2010/0262117	12/2009	Magni et al.	N/A	N/A
2010/0262434	12/2009	Shaya	N/A	N/A
2010/0286997	12/2009	Srinivasan	N/A	N/A
2010/0295686	12/2009	Sloan et al.	N/A	N/A
2010/0298765	12/2009	Budiman et al.	N/A	N/A
2010/0317951	12/2009	Rutkowski et al.	N/A	N/A
2011/0021584	12/2010	Berggren et al.	N/A	N/A
2011/0028817	12/2010	Jin et al.	N/A	N/A
2011/0054390	12/2010	Searle et al.	N/A	N/A
2011/0054399	12/2010	Chong et al.	N/A	N/A
2011/0071765	12/2010	Yodfat et al.	N/A	N/A
2011/0124996	12/2010	Reinke et al.	N/A	N/A
2011/0142688	12/2010	Chappel et al.	N/A	N/A
2011/0144586	12/2010	Michaud et al.	N/A	N/A
2011/0152658	12/2010	Peyser et al.	N/A	N/A
2011/0160652	12/2010	Yodfat et al.	N/A	N/A
2011/0178472	12/2010	Cabiri	N/A	N/A
2011/0190694	12/2010	Lanier, Jr. et al.	N/A	N/A
2011/0193704	12/2010	Harper et al.	N/A	N/A
2011/0202005	12/2010	Yodfat et al.	N/A	N/A
2011/0213306	12/2010	Hanson et al.	N/A	N/A
2011/0218495	12/2010	Remebe	N/A	N/A
2011/0225024	12/2010	Seyer et al.	N/A	N/A
2011/0230833	12/2010	Andman et al.	N/A	N/A
2011/0246235	12/2010	Powell et al.	N/A	N/A
2011/0251509	12/2010	Beyhan et al.	N/A	N/A
2011/0289497	12/2010	Kiaie et al.	N/A	N/A
2011/0313680	12/2010	Doyle, III	N/A	N/A
2011/0316562	12/2010	Cefai et al.	N/A	N/A
2012/0003935	12/2011	Ydon et al.	N/A	N/A
2012/0010594	12/2011	Holt et al.	N/A	N/A
2012/0029941	12/2011	Malave et al.	N/A	N/A
2012/0030393	12/2011	Ganesh et al.	N/A	N/A
2012/0050046	12/2011	Satorius et al.	N/A	N/A
2012/0053556	12/2011	Lee	N/A	N/A
2012/0054841	12/2011	Schultz et al.	N/A	N/A
2012/0078067	12/2011	Kovatchev et al.	N/A	N/A
2012/0078161	12/2011	Masterson et al.	N/A	N/A
2012/0078181	12/2011	Smith et al.	N/A	N/A
2012/0095316	12/2011	Lewis et al.	N/A	N/A
2012/0101451	12/2011	Boit et al.	N/A	N/A
2012/0123234	12/2011	Atlas et al.	N/A	N/A
2012/0136336	12/2011	Mastrototaro et al.	N/A	N/A
2012/0153936	12/2011	Romani et al.	N/A	N/A
2012/0182939	12/2011	Rajan et al.	N/A	N/A
2012/0184909	12/2011	Gyrn	N/A	N/A
2012/0190955	12/2011	Rao et al.	N/A	N/A
2012/0201048	12/2011	Prais	N/A	N/A
2012/0203085	12/2011	Rebec	N/A	N/A
2012/0203178	12/2011	Tverskoy	N/A	N/A
2012/0215087	12/2011	Cobelli et al.	N/A	N/A
2012/0225134	12/2011	Komorowski	N/A	N/A
2012/0226259	12/2011	Yodfat et al.	N/A	N/A
2012/0232520	12/2011	Sloan et al.	N/A	N/A

2012/0238851	12/2011	Kamen et al.	N/A	N/A
2012/0265166	12/2011	Yodfat	N/A	N/A
2012/0271655	12/2011	Knobel et al.	N/A	N/A
2012/0277667	12/2011	Yodat et al.	N/A	N/A
2012/0277668	12/2011	Chawla	N/A	N/A
2012/0282111	12/2011	Nip et al.	N/A	N/A
2012/0295550	12/2011	Wilson et al.	N/A	N/A
2013/0030358	12/2012	Yodfat et al.	N/A	N/A
2013/0030841	12/2012	Bergstrom et al.	N/A	N/A
2013/0036100	12/2012	Nagpal et al.	N/A	N/A
2013/0060194	12/2012	Rostein	N/A	N/A
2013/0080832	12/2012	Dean et al.	N/A	N/A
2013/0138452	12/2012	Cork et al.	N/A	N/A
2013/0158503	12/2012	Kanderian, Jr. et al.	N/A	N/A
2013/0173473	12/2012	Birtwhistle et al.	N/A	N/A
2013/0178791	12/2012	Javitt	N/A	N/A
2013/0204130	12/2012	Mcarthur et al.	N/A	N/A
2013/0231642	12/2012	Doyle et al.	N/A	N/A
2013/0245545	12/2012	Arnold et al.	N/A	N/A
2013/0253472	12/2012	Cabiri	N/A	N/A
2013/0261406	12/2012	Rebec et al.	N/A	N/A
2013/0274576	12/2012	Amirouche et al.	N/A	N/A
2013/0296823	12/2012	Melker et al.	N/A	N/A
2013/0317753	12/2012	Kamen et al.	N/A	N/A
2013/0338576	12/2012	OConnor et al.	N/A	N/A
2014/0005633	12/2013	Finan	N/A	N/A
2014/0012119	12/2013	Geaghan et al.	N/A	N/A
2014/0054883	12/2013	Lanigan et al.	N/A	N/A
2014/0066886	12/2013	Roy et al.	N/A	N/A
2014/0074033	12/2013	Sonderegger et al.	N/A	N/A
2014/0078263	12/2013	Kim	N/A	N/A
2014/0114277	12/2013	Eggert et al.	N/A	N/A
2014/0121635	12/2013	Hayter	N/A	N/A
2014/0128839	12/2013	Dilanni et al.	N/A	N/A
2014/0131199	12/2013	Simmons et al.	N/A	N/A
2014/0135880	12/2013	Baumgartner et al.	N/A	N/A
2014/0146202	12/2013	Boss et al.	N/A	N/A
2014/0148784	12/2013	Anderson et al.	N/A	N/A
2014/0163664	12/2013	Goldsmith	N/A	N/A
2014/0180203	12/2013	Budiman et al.	N/A	N/A
2014/0180240	12/2013	Finan et al.	N/A	N/A
2014/0200426	12/2013	Taub et al.	N/A	N/A
2014/0200559	12/2013	Doyle et al.	N/A	N/A
2014/0230021	12/2013	Birthwhistle et al.	N/A	N/A
2014/0254170	12/2013	Celentano et al.	N/A	N/A
2014/0276554	12/2013	Finan et al.	N/A	N/A
2014/0276556	12/2013	Saint et al.	N/A	N/A
2014/0278123	12/2013	Prodhom et al.	N/A	N/A
2014/0296787	12/2013	Agard et al.	N/A	N/A
2014/0309615	12/2013	Mazlish	N/A	N/A
2014/0316379	12/2013	Sonderegger et al.	N/A	N/A
2014/0325065	12/2013	Birtwhistle et al.	N/A	N/A
2015/0018633	12/2014	Kovachev et al.	N/A	N/A
2015/0025329	12/2014	Amarasingham et al.	N/A	N/A
2015/0025495	12/2014	Peyser	N/A	N/A
2015/0038898	12/2014	Palmer et al.	N/A	N/A
2015/0057913	12/2014	Benhammou	N/A	N/A
2015/0119666	12/2014	Brister et al.	N/A	N/A
2015/0120317	12/2014	Mayou et al.	N/A	N/A
2015/0134265	12/2014	Kohlbrecher et al.	N/A	N/A
2015/0165119	12/2014	Palerm et al.	N/A	N/A
2015/0173674	12/2014	Hayes et al.	N/A	N/A
2015/0213217	12/2014	Amarasingham et al.	N/A	N/A

2015/0217052	12/2014	Keenan et al.	N/A	N/A
2015/0217053	12/2014	Booth et al.	N/A	N/A
2015/0265767	12/2014	Vazquez et al.	N/A	N/A
2015/0283335	12/2014	Lin	N/A	N/A
2015/0290391	12/2014	Schmid et al.	N/A	N/A
2015/0306314	12/2014	Doyle et al.	N/A	N/A
2015/0338349	12/2014	Carter et al.	N/A	N/A
2015/0351671	12/2014	Vanslyke et al.	N/A	N/A
2015/0361154	12/2014	Jowett et al.	N/A	N/A
2015/0366945	12/2014	Greene	N/A	N/A
2016/0015891	12/2015	Papiorek	N/A	N/A
2016/0022905	12/2015	Nagar et al.	N/A	N/A
2016/0038673	12/2015	Morales	N/A	N/A
2016/0038689	12/2015	Lee et al.	N/A	N/A
2016/0051749	12/2015	Istoc	N/A	N/A
2016/0058941	12/2015	Wu et al.	N/A	N/A
2016/0082187	12/2015	Schaible et al.	N/A	N/A
2016/0089494	12/2015	Guerrini	N/A	N/A
2016/0135747	12/2015	Frey et al.	N/A	N/A
2016/0175520	12/2015	Palerm et al.	N/A	N/A
2016/0184517	12/2015	Baek et al.	N/A	N/A
2016/0228641	12/2015	Gescheit et al.	N/A	N/A
2016/0243318	12/2015	Despa et al.	N/A	N/A
2016/0256087	12/2015	Doyle et al.	N/A	N/A
2016/0287512	12/2015	Cooper et al.	N/A	N/A
2016/0302054	12/2015	Kimura et al.	N/A	N/A
2016/0310665	12/2015	Hwang et al.	N/A	N/A
2016/0331310	12/2015	Kovatchev	N/A	N/A
2016/0339172	12/2015	Michaud et al.	N/A	N/A
2016/0354543	12/2015	Cinar et al.	N/A	N/A
2017/0028132	12/2016	Cronenberg et al.	N/A	N/A
2017/0049386	12/2016	Abraham et al.	N/A	N/A
2017/0106138	12/2016	Cabiri	N/A	N/A
2017/0143899	12/2016	Gondhalekar et al.	N/A	N/A
2017/0143900	12/2016	Rioux et al.	N/A	N/A
2017/0156682	12/2016	Doyle et al.	N/A	N/A
2017/0173261	12/2016	O'Connor et al.	N/A	N/A
2017/0189270	12/2016	Nazzaro et al.	N/A	N/A
2017/0189625	12/2016	Cirillo et al.	N/A	N/A
2017/0214584	12/2016	Kanojia et al.	N/A	N/A
2017/0234858	12/2016	Depa et al.	N/A	N/A
2017/0281877	12/2016	Marlin et al.	N/A	N/A
2017/0296746	12/2016	Chen et al.	N/A	N/A
2017/0311903	12/2016	Davis et al.	N/A	N/A
2017/0348479	12/2016	Choate et al.	N/A	N/A
2017/0348482	12/2016	Duke et al.	N/A	N/A
2017/0354785	12/2016	Gazeley et al.	N/A	N/A
2018/0015274	12/2017	Haury et al.	N/A	N/A
2018/0036495	12/2017	Searle et al.	N/A	N/A
2018/0040255	12/2017	Freeman et al.	N/A	N/A
2018/0055452	12/2017	Breton	N/A	N/A
2018/0075200	12/2017	Davis et al.	N/A	N/A
2018/0075201	12/2017	Davis et al.	N/A	N/A
2018/0075202	12/2017	Davis et al.	N/A	N/A
2018/0092576	12/2017	O'Connor et al.	N/A	N/A
2018/0126073	12/2017	Wu et al.	N/A	N/A
2018/0169334	12/2017	Grosman et al.	N/A	N/A
2018/0200434	12/2017	Mazlish et al.	N/A	N/A
2018/0200438	12/2017	Mazlish et al.	N/A	N/A
2018/0200441	12/2017	Desborough et al.	N/A	N/A
2018/0204636	12/2017	Edwards et al.	N/A	N/A
2018/0207357	12/2017	John	N/A	N/A
2018/0236173	12/2017	Mccaffrey et al.	N/A	N/A

2018/0256815	12/2017	Nazzaro	N/A	N/A
2018/0277253	12/2017	Gondhalekar et al.	N/A	N/A
2018/0280609	12/2017	Nishimura et al.	N/A	N/A
2018/0289891	12/2017	Finan et al.	N/A	N/A
2018/0296757	12/2017	Finan et al.	N/A	N/A
2018/0307515	12/2017	Meller et al.	N/A	N/A
2018/0342317	12/2017	Skirble et al.	N/A	N/A
2018/0369479	12/2017	Hayter et al.	N/A	N/A
2019/0022317	12/2018	Uddin et al.	N/A	N/A
2019/0076600	12/2018	Grosman et al.	N/A	N/A
2019/0091404	12/2018	Nazzaro et al.	N/A	N/A
2019/0132801	12/2018	Kamath et al.	N/A	N/A
2019/0167895	12/2018	Dechellette et al.	N/A	N/A
2019/0240403	12/2018	Palerm et al.	N/A	N/A
2019/0240417	12/2018	Hostettler et al.	N/A	N/A
2019/0290844	12/2018	Monirabbasi et al.	N/A	N/A
2019/0321545	12/2018	Saint	N/A	N/A
2019/0336683	12/2018	O'Connor et al.	N/A	N/A
2019/0336684	12/2018	O'Connor et al.	N/A	N/A
2019/0348157	12/2018	Booth et al.	N/A	N/A
2020/0046268	12/2019	Patek et al.	N/A	N/A
2020/0101222	12/2019	Intereur et al.	N/A	N/A
2020/0101223	12/2019	Lintereur et al.	N/A	N/A
2020/0101225	12/2019	O'Connor et al.	N/A	N/A
2020/0197605	12/2019	Haidar	N/A	N/A
2020/0219625	12/2019	Kahlbaugh	N/A	N/A
2020/0261643	12/2019	Boyaval et al.	N/A	N/A
2020/0342974	12/2019	Chen et al.	N/A	N/A
2021/0050085	12/2020	Hayter et al.	N/A	N/A
2021/0098105	12/2020	Lee et al.	N/A	N/A
2021/0256872	12/2020	Matsumoto	N/A	G16H 50/20
2022/0023536	12/2021	Graham et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2015200834	12/2014	AU	N/A
2015301146	12/2016	AU	N/A
2863379	12/2012	CA	N/A
1297140	12/2000	CN	N/A
201134101	12/2007	CN	N/A
19756872	12/1998	DE	N/A
0341049	12/1988	EP	N/A
0496305	12/1991	EP	N/A
0549341	12/1992	EP	N/A
1491144	12/2003	EP	N/A
0801578	12/2005	EP	N/A
1762263	12/2006	EP	N/A
1839694	12/2006	EP	N/A
1852703	12/2006	EP	N/A
2099384	12/2008	EP	N/A
2139382	12/2009	EP	N/A
2353628	12/2010	EP	N/A
2397181	12/2010	EP	N/A
2666520	12/2012	EP	N/A
2695573	12/2013	EP	N/A
1874390	12/2013	EP	N/A
2830499	12/2014	EP	N/A
2943149	12/2014	EP	N/A
3068290	12/2015	EP	N/A
3135965	12/2016	EP	N/A
3177344	12/2016	EP	N/A
3187201	12/2016	EP	N/A
3314548	12/2017	EP	N/A

3000497	12/2018	EP	N/A
1571582	12/2018	EP	N/A
2897071	12/2018	EP	N/A
3598942	12/2019	EP	N/A
3607985	12/2019	EP	N/A
2559866	12/2015	ES	N/A
2096275	12/1971	FR	N/A
357139	12/1930	GB	N/A
810488	12/1958	GB	N/A
1401588	12/1974	GB	N/A
2176595	12/1985	GB	N/A
2443260	12/2007	GB	N/A
2443261	12/2007	GB	N/A
2461086	12/2008	GB	N/A
2495014	12/2012	GB	N/A
2524717	12/2014	GB	N/A
2525149	12/2014	GB	N/A
51125993	12/1975	JP	N/A
02131777	12/1989	JP	N/A
2001190659	12/2000	JP	N/A
2003154190	12/2002	JP	N/A
2007144141	12/2006	JP	N/A
2004283378	12/2006	JP	N/A
2007307359	12/2006	JP	N/A
2008242502	12/2007	JP	N/A
2009523535	12/2008	JP	N/A
2012210441	12/2011	JP	N/A
2017525451	12/2016	JP	N/A
2018153569	12/2017	JP	N/A
2019525276	12/2018	JP	N/A
2008024814	12/2007	NO	N/A
200740148	12/2006	TW	N/A
M452390	12/2012	TW	N/A
9800193	12/1997	WO	N/A
9801071	12/1997	WO	N/A
9819145	12/1997	WO	N/A
9824495	12/1997	WO	N/A
9841267	12/1997	WO	N/A
9956803	12/1998	WO	N/A
0010628	12/1999	WO	N/A
0013580	12/1999	WO	N/A
0019887	12/1999	WO	N/A
0030705	12/1999	WO	N/A
200032258	12/1999	WO	N/A
0061215	12/1999	WO	N/A
0078210	12/1999	WO	N/A
0172354	12/2000	WO	N/A
2002015954	12/2001	WO	N/A
2002043866	12/2001	WO	N/A
2002082990	12/2001	WO	N/A
2003016882	12/2002	WO	N/A
2003039362	12/2002	WO	N/A
2003045233	12/2002	WO	N/A
2004043250	12/2003	WO	N/A
2005110601	12/2003	WO	N/A
2004092715	12/2003	WO	N/A
2005031631	12/2004	WO	N/A
2005051170	12/2004	WO	N/A
2005082436	12/2004	WO	N/A
2005113036	12/2004	WO	N/A
2006053007	12/2005	WO	N/A
2006060668	12/2005	WO	N/A
2007064835	12/2006	WO	N/A

2007078937	12/2006	WO	N/A
2007084214	12/2006	WO	N/A
2007092618	12/2006	WO	N/A
2007112034	12/2006	WO	N/A
2008024810	12/2007	WO	N/A
2008029403	12/2007	WO	N/A
2008133702	12/2007	WO	N/A
2009023634	12/2008	WO	N/A
2009032399	12/2008	WO	N/A
2009045462	12/2008	WO	N/A
2009049252	12/2008	WO	N/A
2009066287	12/2008	WO	N/A
2009066288	12/2008	WO	N/A
2009098648	12/2008	WO	N/A
2009134380	12/2008	WO	N/A
2010025433	12/2009	WO	N/A
2010053702	12/2009	WO	N/A
2010078434	12/2009	WO	N/A
2010132077	12/2009	WO	N/A
2010138848	12/2009	WO	N/A
2010146579	12/2009	WO	N/A
2010147659	12/2009	WO	N/A
2011012465	12/2010	WO	N/A
2011095483	12/2010	WO	N/A
2011133823	12/2010	WO	N/A
2012045667	12/2011	WO	N/A
2012108959	12/2011	WO	N/A
2012134588	12/2011	WO	N/A
2012177353	12/2011	WO	N/A
2012178134	12/2011	WO	N/A
2013078200	12/2012	WO	N/A
2013134486	12/2012	WO	N/A
2013149186	12/2012	WO	N/A
20130149186	12/2012	WO	N/A
2013177565	12/2012	WO	N/A
2013182321	12/2012	WO	N/A
2014109898	12/2013	WO	N/A
2014110538	12/2013	WO	N/A
2014136105	12/2013	WO	N/A
2014194183	12/2013	WO	N/A
2015056259	12/2014	WO	N/A
2015061493	12/2014	WO	N/A
2015073211	12/2014	WO	N/A
2015081337	12/2014	WO	N/A
2015187366	12/2014	WO	N/A
2015187793	12/2014	WO	N/A
2016004088	12/2015	WO	N/A
2016022650	12/2015	WO	N/A
2016041873	12/2015	WO	N/A
2016089702	12/2015	WO	N/A
2016141082	12/2015	WO	N/A
2016161254	12/2015	WO	N/A
2016181384	12/2015	WO	N/A
2017004278	12/2016	WO	N/A
2017089289	12/2016	WO	N/A
2017091624	12/2016	WO	N/A
2017105600	12/2016	WO	N/A
2017184988	12/2016	WO	N/A
2017205816	12/2016	WO	N/A
2018009614	12/2017	WO	N/A
2018067748	12/2017	WO	N/A
2018120104	12/2017	WO	N/A
2018136799	12/2017	WO	N/A

2018204568	12/2017	WO	N/A
2019043702	12/2018	WO	N/A
2019077482	12/2018	WO	N/A
2019094440	12/2018	WO	N/A
2019195521	12/2018	WO	N/A
2019213493	12/2018	WO	N/A
2019246381	12/2018	WO	N/A
2020081393	12/2019	WO	N/A
2020124058	12/2019	WO	N/A
2021011738	12/2020	WO	N/A

OTHER PUBLICATIONS

US 5,954,699 A, 09/1999, Jost et al. (withdrawn) cited by applicant

Breton et al., "Continuous Glucose Monitoring and Insulin Informed Advisory System with Automated Titration and Dosing of Insulin Reduces Glucose Variability in Type 1 Diabetes Mellitus", Diabetes Technology & Therapeutics, vol. 20, No. 8, 2018. (Year: 2018). cited by examiner

European Search Report for the European Patent Application No. EP03743667, dated Jul. 22, 2008. cited by applicant

International Search Report and Written Opinion mailed Sep. 9, 2016, issued in PCT Patent Application No. PCT/US2016/037189, 12 pages. cited by applicant

Preliminary Report on Patentability mailed Dec. 21, 2017, issued in PCT Patent Application No. PCT/US2016/037189. cited by applicant

U.K. Intellectual Property Office, GB Application No. GB 1401587.9, "Search Report under Section 17(5)" Aug. 11, 2015, 1 page. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/GB2015/050247, May 8, 2015, 14 pages. cited by applicant

Extended Search Report mailed Nov. 24, 2017, issued in European Patent Application No. 15779465.2, 10 pages. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/US15/26875, mailed Jan. 18, 2016, 10 pages. cited by applicant

U.K. Intellectual Property Office, GB Application No. GB 1401588.7, "Search Report under Section 17(5)" Aug. 17, 2015, 1 page. cited by applicant

U.K. Intellectual Property Office, GB Application No. GB 1401589.5, "Search Report under Section 17" Jul. 27, 2015, 1 page. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/GB2015/050250, May 7, 2015, 9 pages. cited by applicant

3GPP TS 23.003 V10.0.0.0 Numbering, addressing and identification. Dec. 2010. cited by applicant

International Search Report and Written Opinion for PCT Application No. PCT/GB2015/050251, Jun. 12, 2015, 9 pages. cited by applicant

European Search Report for the European Patent Application No. EP19194241, dated Oct. 22, 2019, 6 pages. cited by applicant

International Preliminary Report on Patentability for PCT/US2017/061095, issued on May 14, 2019, 6 pages. cited by applicant

International Search Report and Written Opinion for PCT/US18/52468, mailed on Feb. 26, 2019, 16 pages. cited by applicant

International Search Report and Written Opinion for PCT/US2017/061095, mailed on Feb. 20, 2018, 8 pages. cited by applicant

Unger, Jeff, et al., "Glucose Control in the Hospitalized Patient," Emerg. Med 36(9):12-18 (2004). cited by applicant

"Glucommander FAQ" downloaded from <https://adaendo.com/GlucommanderFAQ.html> on Mar. 16, 2009. cited by applicant

Finfer, Simon & Heritier, Stephane. (2009). The NICE-SUGAR (Normoglycaemia in Intensive Care Evaluation and Survival Using Glucose Algorithm Regulation) Study: statistical analysis plan. Critical care and resuscitation : journal of the Australasian Academy of Critical Care Medicine. 11. 46-57. cited by applicant

Letters to the Editor regarding "Glucose Control in Critically Ill Patients," N Engl J Med 361: 1, Jul. 2, 2009. cited by applicant

"Medtronic is Leading a Highly Attractive Growth Market," Jun. 2, 2009. cited by applicant

Davidson, Paul C., et al. "Glucommander: An Adaptive, Computer-Directed System for IV Insulin Shown to be Safe, Simple, and Effective in 120,618 Hours of Operation," Atlanta Diabetes Associates presentation. cited by applicant

Davidson, Paul C., et al. "Pumpmaster and Glucommander," presented at the MiniMed Symposium, Atlanta GA, Dec. 13, 2003. cited by applicant

Kanji S., et al. "Reliability of point-of-care testing for glucose measurement in critically ill adults," Critical Care Med, vol. 33, No. 12, pp. 2778-2785, 2005. cited by applicant

Krinsley James S., "Severe hypoglycemia in critically ill patients: Risk factors and outcomes," Critical Care Med, vol. 35, No. 10, pp. 1-6, 2007. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/016283, mailed Jun. 2, 2021, 15 pages. cited by applicant

Farkas et al. "Single-Versus Triple-Lumen Central Catheter-Related Sepsis: A Prospective Randomized Study in a Critically Ill Population" The American Journal of Medicine, Sep. 1992, vol. 93, p. 277-282. cited by applicant

Davidson, Paul C., et al., A computer-directed intravenous insulin system shown to be safe, simple, and effective in 120,618 h of operation, Diabetes Care, vol. 28, No. 10, Oct. 2005, pp. 2418-2423. cited by applicant

R Anthony Shaw, et al., "Infrared Spectroscopy in Clinical and Diagnostic Analysis," Encyclopedia of Analytical Chemistry, ed.

Robert A. Meyers, John Wiley & Sons, Ltd., pp. 1-20, 2006. cited by applicant

Gorke, A “Microbial Contamination Of Haemodialysis Catheter Connections” Journal of Renal Care, European Dialysis & Transplant Nurses Association. cited by applicant

Lovich et al. “Central venous catheter infusions: A laboratory model shows large differences in drug delivery dynamics related to catheter dead volume” Critical Care Med 2007 vol. 35, No. 12. cited by applicant

Van Den Berghe, Greet, M.D., Ph.D., et al., Intensive Insulin Therapy in Critically Ill Patients, The New England Journal of Medicine, vol. 345, No. 19, Nov. 8, 2001, pp. 1359-1367. cited by applicant

Schlegel et al., “Multilumen Central Venous Catheters Increase Risk for Catheter-Related Bloodstream Infection: Prospective Surveillance Study” Infection 2008; 36: 322-327. cited by applicant

Wilson, George S., et al., Progress toward the Development of an Implantable Sensor for Glucose, Clin. Chem., vol. 38, No. 9, 1992 pp. 1613-1617. cited by applicant

Yeung et al. “Infection Rate for Single Lumen v Triple Lumen Subclavian Catheters” Infection Control and Hospital Epidemiology, vol. 9, No. 4 (Apr. 1988) pp. 154-158 The University of Chicago Press. cited by applicant

International Search Report and Written Opinion, International Application No. PCT/US2010/033794 mailed Jul. 16, 2010. cited by applicant

International Search Report and Written Opinion in PCT/US2008/079641 dated Feb. 25, 2009. cited by applicant

Berger, “Measurement of Analytes in Human Serum and Whole Blood Samples by Near-Infrared Raman Spectroscopy,” Ph.D. Thesis, Massachusetts Institute of Technology, Chapter 4, pp. 50-73, 1998. cited by applicant

Berger, “An Enhanced Algorithm for Linear Multivariate Calibration,” Analytical Chemistry, vol. 70, No. 3, pp. 623-627, Feb. 1, 1998. cited by applicant

Billman et al., “Clinical Performance of an In line Ex-Vivo Point of Care Monitor: A Multicenter Study,” Clinical Chemistry 48: 11, pp. 2030-2043, 2002. cited by applicant

Widness et al., “Clinical Performance on an In-Line Point-of-Care Monitor in Neonates”; Pediatrics, vol. 106, No. 3, pp. 497-504, Sep. 2000. cited by applicant

Finkielman et al., “Agreement Between Bedside Blood and Plasma Glucose Measurement in the ICU Setting”; retrieved from <http://www.chestjournal.org>; CHEST/127/5/May 2005. cited by applicant

Glucon Critical Care Blood Glucose Monitor; Glucon; retrieved from <http://www.glucon.com>. cited by applicant

Fogt, et al., “Development and Evaluation of a Glucose Analyzer for a Glucose-Controlled Insulin Infusion System (Biostator)”; Clinical Chemistry, vol. 24, No. 8, pp. 1366-1372, 1978. cited by applicant

Vonach et al., “Application of Mid-Infrared Transmission Spectrometry to the Direct Determination of Glucose in Whole Blood,” Applied Spectroscopy, vol. 52, No. 6, 1998, pp. 820-822. cited by applicant

Muniyappa et al., “Current Approaches for assessing insulin sensitivity and resistance in vivo: advantages, imitations, and appropriate usage,” AJP-Endocrinol Metab, vol. 294, E15-E26, first published Oct. 23, 2007. cited by applicant

International Preliminary Report on Patentability for the International Patent Application No. PCT/US2019/053603, mailed Apr. 8, 2021, 9 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2019/053603, mailed Jan. 7, 2020, 16 pages. cited by applicant

Dassau et al., “Detection of a meal using continuous glucose monitoring: Implications for an artificial [beta]-cell.” Diabetes Care, American Diabetes Association, Alexandria, VA, US, 31(2):295-300 (2008). cited by applicant

Cameron et al., “Probabilistic Evolving Meal Detection and Estimation of Meal Total Glucose Appearance Author Affiliations”, J Diabetes Sci and Tech, vol. Diabetes Technology Society ;(5):1022-1030 (2009). cited by applicant

Lee et al., “A closed-loop artificial pancreas based on model predictive control: Human-friendly identification and automatic meal disturbance rejection”, Biomedical Signal Processing and Control, Elsevier, Amsterdam, NL, 4(4):1746-8094 (2009). cited by applicant

Anonymous: “Fuzzy control system”, Wikipedia, Jan. 10, 2020. URL: https://en.wikipedia.org/w/index.php?title=Fuzzy_control_system&oldid=935091190. cited by applicant

An Emilia Fushimi: “Artificial Pancreas: Evaluating the ARG Algorithm Without Meal Annoucement”, Journal of Diabetes Science and Technology Diabetes Technology Society, Mar. 22, 2019, pp. 1025-1043. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/017441, mailed May 25, 2021, 12 pages. cited by applicant

Mirko Messori et al: “Individualized model predictive control for the artificial pancreas: In silico evaluation of closed-loop glucose control”, IEEE Control Systems, vol. 38, No. 1, Feb. 1, 2018, pp. 86-104. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/017662, mailed May 26, 2021, 14 pages. cited by applicant

Anonymous: “Reservoir Best Practice and Top Tips” Feb. 7, 2016, URL: <https://www.medtronic-diabetes.co.uk/blog/reservoir-best-practice-and-top-tips>, p. 1. cited by applicant

Gildon Bradford: “InPen Smart Insulin Pen System: Product Review and User Experience” Diabetes Spectrum, vol. 31, No. 4, Nov. 15, 2018, pp. 354-358. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/016050, mailed May 27, 2021, 16 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2020/065226, mailed May 31, 2021, 18 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/017659, mailed May 31,

2021, 13 pages. cited by applicant

Montaser Eslam et al., "Seasonal Local Models for Glucose Prediction in Type 1 Diabetes", IEE Journal of Biomedical and Health Informatics, IEEE, Piscataway, NJ, USA, vol. 24, No. 7, Nov. 29, 2019, pp. 2064-2072. cited by applicant

Samadi Sediqeh et al., "Automatic Detection and Estimation of Unannounced Meals for Multivariable Artificial Pancreas System", Diabetes Technology & Therapeutics, vol. 20m No. 3, Mar. 1, 2018, pp. 235-246. cited by applicant

Samadi Sediqeh et al., "Meal Detection and Carbohydrate Estimation Using Continuous Glucose Sensor Data" IEEE Journal of Biomedical and Health Informatics, IEEE, Piscataway, NJ, USA, vol. 21, No. 3, May 1, 2017, pp. 619-627. cited by applicant

Khodaei et al., "Physiological Closed-Loop Control (PCLC) Systems: Review of a Modern Frontier in Automation", IEEE Access, IEEE, USA, vol. 8, Jan. 20, 2020, pp. 23965-24005. cited by applicant

E. Atlas et al., "MD-Logic Artificial Pancreas System: A pilot study in adults with type 1 diabetes", Diabetes Care, vol. 33, No. 5, Feb. 11, 2010, pp. 1071-1076. cited by applicant

International Preliminary Report on Patentability for the International Patent Application No. PCT/US2019/042160, mailed Jan. 28, 2021, 12 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/047695, mailed Jan. 31, 2022, 26 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US21/064056, mailed Apr. 4, 2022, 12 pages. cited by applicant

Anonymous: "AndroidAPS ComponentOverview", AndroidAPS documentation, Nov. 12, 2020 (Nov. 12, 2020), pp. 1-7, Retrieved from the Internet:
URL:<https://github.com/openaps/AndroidAPSdocs/blob/199ef86a900adf4b3d9c32f605eb11047bd3d62f/docs/EN/Module/module.r>
[retrieved on Apr. 11, 2022] the whole document. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US22/018700, mailed Jun. 7, 2022, 13 pages. cited by applicant

Legacy Med Search, Insulet Enrolls First Patients in Clinical Trial for Omnipod, Sep. 16, 2016, available at URL:
<https://legacymedsearch.com/insulet-enrolls-first-patients-in-clinical-trial-for-omnipod-artificial-pancreas-system/>. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2020/052125, mailed Aug. 12, 2020, 15 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2020/050332, mailed Sep. 12, 2020, 12 pages. cited by applicant

European Patent Office, "Notification of Transmittal of the ISR and the Written Opinion of the International Searching Authority, or the Declaration," in PCT Application No. PCT/GB2015/050248, Jun. 23, 2015, 12 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/012246, mailed Apr. 13, 2021, 15 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/013639, mailed Apr. 28, 2021, 14 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2020/063326, mailed May 3, 2021, 17 pages. cited by applicant

European Search Report for the European Patent Application No. 21168591, mailed Oct. 13, 2021, 4 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/041954, mailed Oct. 25, 2021, 13 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/022694, mailed Jun. 25, 2021, 13 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/017664, mailed May 26, 2021, 16 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2022/012896, mailed Apr. 22, 2022, 15 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2022/013470, mailed May 6, 2022, 14 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2022/013473, mailed May 6, 2022, 13 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2022/019079, mailed Jun. 2, 2022, 14 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2022/018453, mailed Jun. 2, 2022, 13 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US22/019080, mailed Jun. 7, 2022, 14 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US22/019664, mailed Jun. 7, 2022, 14 pages. cited by applicant

International Search Report and Written Opinion for International Patent Application No. PCT/US2021/051027, mailed on Jan. 7, 2022, 16 pages. cited by applicant

International Search Report and Written Opinion for International Patent Application No. PCT/US2021/052372, mailed Jan. 26, 2022, 15 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/046607, mailed Jan. 31,

2022, 20 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/055745, mailed Feb. 14, 2022, 13 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US21/060618, mailed Mar. 21, 2022, 15 pages. cited by applicant

Herrero Pau et al: "Enhancing automatic closed-loop glucose control in type 1 diabetes with an adaptive meal bolus calculator -in silico evaluation under intra-day variability", Computer Methods and Programs in Biomedicine, Elsevier, Amsterdam, NL, vol. 146, Jun. 1, 2017 (Jun. 1, 2017), pp. 125-131, XP085115607, ISSN: 0169-2607, DOI:10.1016/J.CMPB.2017.05.010. cited by applicant

Marie Aude Qemerais: "Preliminary Evaluation of a New Semi-Closed-Loop Insulin Therapy System over the prandial period in Adult Patients with type I diabetes: the WP6. 0 Diabeloop Study", Journal of Diabetes Science and Technology Diabetes Technology Society Reprints and permissions, Jan. 1, 2014, pp. 1177-1184, Retrieved from the Internet: URL:<http://journals.sagepub.com/doi/pdf/10.1177/1932296814545668> [retrieved on Jun. 6, 2022] chapter "Functioning of the Algorithm" chapter "Statistical Analysis" p. 1183, left-hand col. line 16- line 23. cited by applicant

Anonymous: "Kernel density estimation", Wikipedia, Nov. 13, 2020 (Nov. 13, 2020), pp. 1-12, XP055895569, Retrieved from the Internet: URL:https://en.wikipedia.org/w/index.php?title=Kernel_density_estimation&oldid=988508333 [retrieved on Jun. 6, 2022] cited by applicant

Anonymous: "openaps / oref0 / lib/determine-basal-js", openaps repository, Nov. 9, 2019 (Nov. 9, 2019), pp. 1-17, XP055900283, Retrieved from the Internet: URL:<https://github.com/openaps/oref0/blob/master/lib/determine-basal/determine-basal.js> [retrieved on Jun. 6, 2022] line 116-line 118, line 439-line 446. cited by applicant

Anonymous: "AndroidAPS screens", AndroidAPS documentation, Oct. 4, 2020 (Oct. 4, 2020), pp. 1-12, XP055894824, Retrieved from the Internet: URL:<https://github.com/openaps/AndroidAPSdocs/blob/25d8acf8b28262b411b34f416f173ac0814d7e14/docs/EN/Getting-Started/Screenshots.md> [retrieved on Jun. 6, 2022]. cited by applicant

Kozak Milos et al: "Issue #2473 of AndroidAPS", MilosKozak / AndroidAPS Public repository, Mar. 4, 2020 (Mar. 4, 2020), pp. 1-4, XP055900328, Retrieved from the Internet: URL:<https://github.com/MilosKozak/AndroidAPS/issues/2473> [retrieved on Jun. 6, 2022]. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/052855, mailed Dec. 22, 2021, 11 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/047771, mailed Dec. 22, 2021, 11 pages. cited by applicant

Medication Bar Code System Implementation Planning Section I: A Bar Code Primer for Leaders, Aug. 2013. cited by applicant

Medication Bar Code System Implementation Planning Section II: Building the Case for Automated Identification of Medications, Aug. 2013. cited by applicant

Villareal et al. (2009) in: Distr. Comp. Art. Intell. Bioninf. Soft Comp. Amb. Ass. Living; Int. Work Conf. Art. Neural Networks (IWANN) 2009, Lect. Notes Comp. Sci. vol. 5518; S. Omatu et al. (Eds.), pp. 870-877. cited by applicant

Fox, Ian G .; Machine Learning for Physiological Time Series: Representing and Controlling Blood Glucose for Diabetes Management; University of Michigan. ProQuest Dissertations Publishing, 2020. 28240142. (Year: 2020). cited by applicant

International Search Report and Written Opinion for International Patent Application No. PCT/US2022/029012, mailed Aug. 19, 2022, 12 pages. cited by applicant

Anonymous: "Artificial pancreas—Wikipedia", Mar. 13, 2018 (Mar. 13, 2018), XP055603712, Retrieved from the Internet: URL: https://en.wikipedia.org/wiki/Artificial_pancreas [retrieved on Jul. 9, 2019] section "Medical Equipment" and the figure labeled "The medical equipment approach to an artificial pancreas". cited by applicant

Kaveh et al., "Blood Glucose Regulation via Double Loop Higher Order Sliding Mode Control and Multiple Sampling Rate." Paper presented at the proceedings of the 17th IFAC World Congress, Seoul, Korea (Jul. 2008). cited by applicant

Dassau et al., "Real-Time Hypoglycemia Prediction Suite Using Continuous Glucose Monitoring," Diabetes Care, vol. 33, No. 6, 1249-1254 (2010). cited by applicant

International Search Report and Written Opinion for International Patent Application No. PCT/US17/53262, mailed on Dec. 13, 2017, 8 pages. cited by applicant

Van Heusden et al., "Control-Relevant Models for Glucose Control using A Priori Patient Characteristics", IEEE Transactions on Biomedical Engineering, vol. 59, No. 7, (Jul. 1, 2012) pp. 1839-1849. cited by applicant

Doyle III et al., "Run-to-Run Control Strategy for Diabetes Management." Paper presented at 23rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society, Istanbul, Turkey. (Oct. 2001). cited by applicant

Bequette, B.W., and Desemone, J., "Intelligent Dosing Systems": Need for Design and Analysis Based on Control Theory, Diabetes Technology and Therapeutics 9(6): 868-873 (2004). cited by applicant

Parker et al., "A Model-Based Algorithm for Blood Glucose Control in Type 1 Diabetic Patients." IEEE Transactions on Biomedical Engineering, 46 (2) 148-147 (1999). cited by applicant

International Search Report and Written Opinion for International Patent Application No. PCT/US2017/015601, mailed May 16, 2017, 12 pages. cited by applicant

International Search Report and Written Opinion for the International Patent Application No. PCT/US2018/018901, mailed on Aug. 6, 2018, 12 pages. cited by applicant

International Search Report and Written Opinion for International Patent Application No. PCT/US2018/052467, mailed Jan. 4, 2019, 13 pages. cited by applicant

"How to Create a QR Code that Deep Links to Your Mobile App", Pure Oxygen Labs, web<<https://pureoxygenlabs.com/how-to->

create-a-qr-codes-that-deep-link-to-your-mobile-app/> Year:2017. cited by applicant
“Read NFC Tags with an iPhone App on iOS 11”, GoToTags, Sep. 11, 2017, web <<https://gototags.com/blog/read-nfc-tags-with-an-iphone-app-on-ios-11/>> (Year:2017). cited by applicant
International Search Report and Written Opinion for International Patent Application No. PCT/US2016/063350, mailed on Mar. 27, 2017, 9 pages. cited by applicant
Extended Search Report mailed Aug. 13, 2018, issued in European Patent Application No. 16753053.4, 9 pages. cited by applicant
International Search Report and Written Opinion for International Patent Application No. PCT/US16/18452, mailed on Apr. 29, 2015, 9 pages. cited by applicant
International Preliminary Report on Patentability mailed Aug. 31, 2017, issued in PCT Patent Application No. PCT/US2016/018452 7 pages. cited by applicant
International Search Report and Written Opinion for International Patent Application No. PCT/US2019/055862, mailed on Mar. 11, 2020. cited by applicant
International Search Report and Written Opinion for Application No. PCT/US2019/030562, Sep. 25, 2019, 19 pages. cited by applicant
International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/047690, mailed Jan. 14, 2022, 13 pages. cited by applicant
International Search Report and Written Opinion for the International Patent Application No. PCT/US2022/055745, mailed Feb. 14, 2022, 13 pages. cited by applicant
International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/053162, mailed Mar. 28, 2022, 18 pages. cited by applicant
International Search Report and Written Opinion for the International Patent Application No. PCT/US2021/064041, mailed Apr. 29, 2022, 11 pages. cited by applicant
International Search Report and Written Opinion for the International Patent Application No. PCT/US2022/015809, mailed Jun. 20, 2022, 15 pages. cited by applicant

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/197,613, filed Jun. 7, 2021, the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

(1) Maintaining healthy lifestyle is both critically important to patients with Type 1 diabetes and challenging at the same time. Many patients have a challenge to maintain their blood glucose level at an appropriate level during or after exercise and may have to take rescue snacks if there is a concern of hypoglycemia. Exercise increases insulin sensitivity and muscle cells are better able to use any available insulin to take up glucose during and after activity. Moreover, during exercise, muscle cells absorb glucose for energy, whether insulin is available or not. As a result, glucose in the blood decreases, leading to potential hypoglycemic events, and causing diabetics to feel fearful of exercise and hypoglycemia due to the over delivery of insulin during or after exercise.

(2) These situations can become more difficult to manage if the patient has a high amount of insulin onboard (JOB) or just had either a post-prandial bolus or a correction bolus. The excess insulin followed by exercise activity may further increase the risk of hypoglycemia as insulin absorption increases from exercise, the exercise itself causes the body to metabolize glucose more quickly, and the additional insulin further acts to reduce blood glucose.

BRIEF SUMMARY

(3) In one aspect, a wearable drug delivery device includes a processor, a pump mechanism, a memory, and a reservoir. The reservoir may contain a liquid drug. The pump mechanism may be communicatively coupled to the processor and fluidically coupled to the reservoir. The memory may be coupled to the processor and operable to store programming code that, when executed by the processor, causes the processor to be operable to receive data indicating physiological conditions of a wearer of the wearable drug delivery device, and evaluate the obtained data with relation to the wearer participating in exercise. The processor may determine whether it is safe for the wearer to participate in exercise based on a result of the evaluation and output an exercise safety signal indicating a result of the determination of whether it is safe or unsafe for the wearer to participate in exercise.

(4) In another aspect, an automatic, wearable drug delivery system may include a drug delivery device and a controller. The drug delivery device may include logic circuitry operable to control the drug delivery device, a reservoir configured to contain a liquid drug, a communication device coupled to the logic circuitry and a pump mechanism responsive to the logic circuitry and fluidically coupled to the reservoir. The controller may include a processor and a memory. The memory may be coupled to the processor and configured to store a medication delivery application, an exercise safety application, and

programming code. When the exercise safety application is executed by the processor, the processor is operable to receive an input from the wearer related to a physiological condition of a wearer. The processor may evaluate data parameters related to physiological conditions of a wearer and the received input related to the physiological condition of a wearer and determine whether the wearer has participated in exercise. The processor, in response to a determination that the wearer has participated in exercise and a result of the evaluation of the data parameters, may generate an exercise safety signal, and present an indication of a result of an interpretation of the exercise safety signal at a user interface. The indication is a safe or unsafe indicator. In response to the generation of the exercise safety signal, the processor may modify a medication treatment plan.

(5) In a further aspect, a non-transitory computer-readable storage medium is provided that may include instructions that, when executed by a processor, cause the processor to determine that a wearer of a wearable drug delivery device is participating in exercise. The instructions may cause the processor to modify an insulin onboard curve in response to the determination that a wearer is participating in exercise and use the modified insulin onboard curve to calculate a next dosage of insulin to be delivered to the wearer and calculate a time to deliver the next dosage. The instructions may trigger the processor to cause the next dosage of insulin to be delivered at the calculated time to the wearer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a flowchart of an example process implemented according to an aspect of the disclosed subject matter.

(2) FIG. 2 is a flowchart of another example process implemented according to a further aspect of the disclosed subject matter.

(3) FIG. 3 is a flowchart of a further example process implemented according to another aspect of the disclosed subject matter.

(4) FIG. 4 illustrates in a flowchart a process in accordance with an embodiment of the disclosed subject matter.

(5) FIG. 5 is a modified insulin onboard decay curve that illustrates an example usable in an embodiment of the disclosed subject matter.

(6) FIG. 6 illustrates an example of a process for generating an exercise model usable with the aspects of the disclosed subject matter.

(7) FIG. 7 illustrates a functional block diagram of a system example suitable for implementing the example processes and techniques described herein.

DETAILED DESCRIPTION

(8) Systems, devices, computer-readable medium and methods in accordance with the present disclosure are now described more fully hereinafter with reference to the accompanying drawings, where one or more embodiments are shown. The systems, devices, and methods may be embodied in many different forms and are not to be construed as being limited to the embodiments set forth herein. Instead, these embodiments are provided so the disclosure is thorough and complete, and may fully convey the scope of methods and devices to those skilled in the art. Each of the systems, devices, and methods disclosed herein provides one or more advantages over conventional systems, components, and methods.

(9) Various examples provide a method, a system, a device and a computer-readable medium for responding to inputs provided by sensors, such as an analyte sensor, and wearers (i.e., users) of an automatic drug delivery system. “Analyte sensor” may refer to a wearable device that includes sensing and measurement devices configured to detect different analytes in a wearer's body and particularly in the user's blood, such as blood glucose levels, and/or other analytes as described herein. The various devices and sensors that may be used to implement some specific examples may also be used to implement different therapeutic regimens using different drugs than those described in the specific examples.

(10) The disclosed examples provide techniques related to the safety of a wearer to participate in exercise that may be used with any additional algorithms or computer applications that manage blood glucose levels and insulin therapy. These algorithms and computer applications may be collectively referred to as “medication delivery algorithms” or “medication delivery applications” and may be operable to deliver different categories of drugs (or medications), such as chemotherapy drugs, pain relief drugs, diabetes treatment drugs (e.g., insulin, glucagon and/or glucagon-like peptides), blood pressure medication, or the like.

(11) A type of medication delivery algorithm (MDA) may include an “artificial pancreas” algorithm-based system, or more generally, an artificial pancreas (AP) application. For ease of discussion, the computer programs and computer applications that implement the medication delivery algorithms or applications may be referred to herein as an “AP application.” An AP application may be configured to provide automatic delivery of insulin based on an analyte sensor input, such as signals received from an analyte sensor, such as a continuous blood glucose monitor, or the like. The signals from the analyte sensor may be blood glucose measurement values, or the like.

(12) In one aspect, a wearable drug delivery device may be operable to determine whether it is safe for a wearer of the drug delivery device to participate in exercise at a present time or future time. The wearable drug delivery device may include, for example, a processor, a reservoir, a pump mechanism, a communication device, and a memory. The reservoir may be configured to contain a liquid drug. The pump mechanism may be communicatively coupled to the processor and fluidically coupled to the reservoir. The memory may be coupled to the processor and configured to store programming code that, when executed by the processor, is operable to perform different functions. The wearable drug delivery device may be communicatively coupled to a controller device. The controller device may have a processor and memory as well as

communication devices that enable communication with the wearable drug delivery device and also external devices and systems, such as cloud-based services. FIG. 1 is a flowchart of an example process implemented according to an aspect of the disclosed subject matter.

(13) The process **100** may be executed by the processor of the controller as the controller processor may have greater computational power than a processor of the wearable drug delivery device. In addition, the controller memory may maintain a greater amount of historical data than the memory of the wearable drug delivery device.

(14) In block **102**, the processor when executing the programming code that implements process **100** may receive data that indicates physiological conditions of a wearer of the wearable drug delivery device. The received data may include a current blood glucose measurement value or data indicative thereof, a total daily insulin value, an amount of a requested bolus, a time at which the bolus was requested, a current insulin onboard value, or an amount of insulin delivered within a past evaluation period. A past evaluation period may be a previous cycle, such as 5 minutes, or another period of time such as a past 15-minute period, a 30-minute period, a 60-minute period, a 120 minute period of time, or a set time period such as when a wearer may most likely exercise, such as between 5 am and 7 am, 12 pm to 1 pm, or the like.

(15) In addition, or alternatively, the processor, when executing the stored programming code, may be further operable to retrieve physiological data from the memory, where the physiological data may contain a number of days of data related to the wearer's participation in exercise. The processor may use the received data, the retrieved physiological data or both to update an exercise model.

(16) In block **104**, the process **100** may evaluate the obtained data with relation to the wearer participating in exercise. For example, the processor, when evaluating the obtained data with relation to the wearer participating in exercise, may be operable to input values obtained from the received data into an exercise model (which may be the updated exercise model), and analyze an output of the exercise model with reference to threshold values related to exercise safety for the wearer. The threshold values related to exercise safety may be parameters that are weighted differently for each wearer according to wearer demographics, disease progression (e.g., type 2 diabetes versus type 1 diabetes), or the like.

(17) In block **106**, the processor implementing process **100** may, based on a result of the evaluation, determine whether it is safe for the wearer to participate in exercise. For example, the processor may be further operable to determine whether the result of the evaluation indicates a presence of one or more conditions and based on the determination of the presence of the one or more conditions. For example, the output of the exercise model may indicate whether it is safe or unsafe for the wearer to exercise and the processor may use the output from the exercise model to make the determination.

(18) In addition, or alternatively, the processor, when determining whether it is safe for the wearer to participate in exercise, may be further operable to determine whether the result of the evaluation indicates that a trajectory of blood glucose measurement values of the wearer is trending downward. Based on the determination that the trajectory of the glucose measurement values of the wearer are trending downward, the processor may generate an indication that exercise is unsafe. For example, the downward trend may be a rate of change of (−) 100 mg/dL per hour for the trajectory of blood glucose measurement values of the wearer. A downward trend may also be referred to as a negative rate of change or a negative (or downward) trajectory of blood glucose measurement values. Of course, rates of change other than (−) 100 mg/dL may be used, such as (−) 10 mg/dL, (−) 50 mg/dL, (−) 75 mg/dL, (−) 90 mg/dL, or the like. Additionally, or alternatively, the output of the exercise model may indicate whether it is safe or unsafe for the wearer to exercise based on a current or future predicted value of the user's blood glucose level. For example, the exercise model may make a prediction based on current blood glucose trends where the user's blood glucose measurement value will be in 30 minutes or 1 hour, and if this predicted value is less than X mg/dL (e.g., where X is approximately 70, 80, 90, 100, or 110 mg/dL), the processor may output an indication that it is unsafe to exercise, or an indication that it is unsafe to exercise for the next Y minutes (for example, where Y is approximately 30 or 60 minutes), or an indication that it is unsafe to exercise unless the user consumes food or carbohydrates, or an indication that it is safe to exercise, or a combination of these potential outputs.

(19) In block **108**, process **100** may cause output of an exercise safety signal indicating a result of the determination of whether it is safe for the wearer to participate in exercise. Based on the determination of whether it safe or unsafe for the wearer to participate in exercise, the processor may generate an indication that exercise is unsafe. Alternatively, the processor may generate that exercise is safe. For example, as described in more detail with reference to other figures, the wearable drug delivery device may also include an output device coupled to the processor. The processor may be further operable to interpret the outputted exercise safety signal. Based on the interpretation, the processor may provide an output signal to the output device based on the outputted exercise safety signal.

(20) For example, the processor may interpret the outputted exercise safety signal as indicating that it is unsafe for the user to exercise. The processor may output command signals that cause the output device to make outputs that are indications of unsafe exercise situations, such as vibrating in a particular pattern of vibrations (e.g., Morse code for distress—S-O-S), make a sound (e.g., a particular beeping sound or number of beeps), change color (e.g., turn from “green” to “yellow” or “red” depending upon the level of urgency (i.e., the degree of unsafety)), or the like. Alternatively, in response to the processor interpreting the outputted exercise safety signal as indicating it is safe to exercise, the processor may output command signals that cause the output device to make outputs that are indications of safe exercise situations, such as vibrating in a particular pattern of vibrations (e.g., 2 vibrations, or the like), make a sound (e.g., a different beeping sound or number of beeps as compared with an unsafe signal), change color (e.g., turn from no lights to a “green” light or from “yellow” to “green”), or the like. Alternatively, if the processor interprets the exercise safety signal as indicating it is safe to exercise, the processor may not generate any output signals.

(21) In addition, or alternatively, the processor may be further operable to modify a medication treatment plan based on the outputted exercise safety signal. For example, if the processor outputs an exercise safety signal indicating it is unsafe to

exercise, but the wearer is determined to participate in exercise, the processor may cause all delivery of insulin to cease for a period of time or reduce an amount of insulin that may be delivered to the user for period of time, included in manual bolus dosages. The cessation or reduction may vary from wearer to wearer.

(22) The determination that the wearer is participating in exercise may be made according to various techniques. For example, the wearer may self-report the participation in exercise by inputting exercise participation in a graphical user interface (provided via an exercise detection application or the AP application) presented on a display of a controller. Alternatively, the controller and/or wearable drug delivery device may have sensors, such as an accelerometer, global positioning system (GPS) signals, Wi-Fi signals, gyroscopes, or the like, that output signals in response to movement by the wearer, a change in location, or an indication of location. Using the information provided by one or more of the respective sensors, the processor when evaluating the information may determine that the information has previously indicated participation in exercise or a location where exercise takes place (e.g., a gym, a ballpark, soccer field, or the like). In addition, or alternatively, the processor may execute an exercise detection application or algorithm that is operable to evaluate inputs from the sensors as well as location information based on the GPS or Wi-Fi signals, information obtained from other mobile computer applications, such as fitness applications, calendar applications (e.g., fitness class schedules, athletic team schedules, or the like). Based on the evaluated inputs, such as high acceleration values for an extended period of time (duration threshold) or a certain number of high acceleration values within another set threshold period of time, a calendar entry indicating the current time as a scheduled time of a fitness class, the processor may determine above a preset probability that the wearer is participating in exercise.

(23) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(24) In another aspect, a wearable drug delivery system may include a drug delivery device, an analyte sensor, and a controller as well as other devices described with reference to other examples. FIG. 2 is a flowchart of another example process implemented according to a further aspect of the disclosed subject matter. The process **200** may be executed by a processor such as the processor of a controller (shown in other examples). For example, the controller may be a smartphone or a dedicated mobile computing device. The processor of the controller may be operable to execute many computer applications including a calendar application, a fitness application, a medication delivery application that utilizes an exercise safety algorithm, such as process **200**, and the like.

(25) In block **202**, the processor may receive an input related to a physiological condition of a wearer. The input may, for example, be from the wearer or from an exercise safety algorithm. In an example, the wearer may request a bolus, use a hypoglycemia protect mode, or indicate that they are about to participate in exercise, or schedule a time at which they will participate in exercise.

(26) In block **204**, in response to the received input related to a physiological condition of a wearer, the process **200** may evaluate data parameters related to one or more physiological conditions of a wearer. The data parameters related to physiological conditions of a wearer may, for example, include a blood glucose measurement value, a total daily insulin amount, a current calculation of an amount of insulin onboard, and an amount of insulin delivered in a past evaluation period.

(27) For example, the processor, when evaluating the data parameters, may be further operable to input data parameters including a blood glucose measurement value, a total daily insulin value, a time and an amount of a requested bolus, an amount of insulin onboard value, or an amount of insulin delivered within a past evaluation period, into an exercise model. The result of the evaluation of the data parameters at block **204** may be an indication that exercise is unsafe or safe at the time of the evaluation. For example, the data parameters may be evaluated utilizing a machine-learning, exercise model that outputs a value indicating whether participation in exercise by a user or wearer is safe or unsafe.

(28) In block **206**, the processor determines whether the wearer has participated in exercise. Participation in exercise may be determined as mentioned above with reference to the exercise determination in FIG. 1. In addition, or alternatively, the determination of exercise participation may be made based on an analysis of different attributes and factors, such as the user's calendar application on the controller, an exercise input entered through a user interface of the controller, an exercise detection application executing on the controller, based on outputs from other sensors on the controller, such as an accelerometer, gyroscope, camera or microphone, or the like. The factors or attributes may be inputs into an exercise model that is operable to determine whether exercise is safe or unsafe based on physiological conditions of a wearer.

(29) In block **208**, in response to a determination that the wearer has participated, or is participating, or will participate in exercise, the processor may generate an exercise safety signal. For example, the processor may use the result of the evaluation of the data parameters obtained in block **204** when generating the exercise safety signal. For example, after participating in exercise, the wearer may indicate that they are feeling “dizzy” or “lightheaded,” and, as a result, the processor can identify (e.g., label) physiological data (e.g., data parameters such as blood glucose measurement values, a total daily insulin amount, a current calculation of an amount of insulin onboard, and an amount of insulin delivered in a past evaluation period) that correspond to the indicated feeling of dizziness or lightheadedness.

(30) Additionally, the processor may be operable to control a communication device to establish a wireless communication connection with an external device, where the external device is a data network device (shown in another example) or a cloud-based service (also shown in another example) to access external devices or services, and retrieve physiological data from a memory source (such as cloud-based storage or the like), where the physiological data contains a number of days of data related to the wearer's participation in exercise. The wearable drug delivery device may also include updates to an exercise model using the retrieved physiological data.

(31) In block **210**, process **200** when executed may cause the processor to present an indication of a result of an

interpretation of the exercise safety signal, wherein the indication is a safe or unsafe indicator for exercise participation. For example, the presentation of the safe or unsafe indicator may be on a display of a controller, such as a touchscreen display, or a user interface of a drug delivery device, or the like.

(32) Alternatively, in a further embodiment, based upon the construction of the exercise model, personalized data and daily blood glucose measurement value trend averages, the processor may suggest the best suitable times for the wearer to participate in exercise. As such, the presented indication may suggest to the wearer to wait 15 minutes, 30 minutes or the like before participating, suggest alternative exercises to perform (e.g., weightlifting (short bursts) versus a long run (an extended high heart rate), or recommend ingesting carbohydrates (e.g., consuming a snack or the like) before exercising.

(33) In addition to predicting safe or unsafe conditions, the exercise model may be further trained to predict a class of probabilities that may be used as threshold values related to exercise safety. In this case, the model may predict which intensity of the exercise may be unsafe. For example, for given physiological conditions of a wearer, high intensity training may not be suitable at this time for the wearer, but a brisk walk or yoga might be acceptable. The recommendations for the type of exercise to participate in and/or to avoid may be presented to the wearer via a user interface of a controller or other device. In addition, or alternatively, a class of probabilities may be selected such that the model favors false positives over false negatives because the consequences of false negatives are higher. This is similar to screening tests where further investigation is needed. In this example, the model may favor a conservative approach where a signal indicating that it is unsafe to exercise is better than the converse. The probabilities could be constructed by retrospect of past glucose distributions under the same activities. For example, if a current blood glucose measurement value is below the 10th percentile of the blood glucose measurement values during past training activities, then the processor executing an exercise safety algorithm may determine that it is not safe for the wearer to do the same activity as in the past training activities at this time.

(34) In block **212**, the processor when executing process **200** may cause a medication delivery application to modify a medication treatment plan in response to a presentation of unsafe indicator. For example, if the processor determines it is unsafe to exercise the processor may withhold delivery of insulin for a period of time. Alternatively, if the processor determines it is safe to proceed with exercise, the processor may cause a modification of a dosage of insulin but continue with the presently set delivery schedule for the next dose.

(35) Either of the steps in block **210** or **212** may be optional depending upon user preference settings or clinician settings or other settings (e.g., default settings). For example, a user preference may be set to notify the user only when exercise is unsafe and no notification when exercise is safe. Other user preference settings may be, for example, to receive notifications whether exercise is safe or unsafe, or receive a notification only when exercise is determined to be safe. It may be helpful to have a couple of exemplary physiological conditions of a wearer as well as other factors that are evaluated by the exercise model.

(36) Each row of Table 1 below may represent a different day for a particular wearer or a group of different wearers. In the table below, the exercise model may evaluate the physiological data related to the physiological conditions of a wearer, such as one or more blood glucose measurement values provided by a continuous glucose monitor (CGM), an indication of whether any blood glucose measurement values were missed, a number of boluses delivered, an amount of insulin (in Units) delivered within the last 30 minutes, whether the wearer's blood glucose measurement value was within the hypoglycemic range (for the wearer, such as, for example, 60 mg/dL or lower), a calculated amount of insulin onboard (IOB) for the wearer, the wearer's target blood glucose setting, and the wearer's TDI. The label is the binary indication of safe to exercise (Label=0) or unsafe to exercise (Label=1).

(37) TABLE-US-00001 TABLE 1 Missed Insulin in intensity CGM count bolus last 30 min of exercise IOB setpoint TDI
label 160 0 0 0 1 0.421 110 45 0 159 0 0 4 1 0.374 110 100 0 158 1 0 10 2 4.332 110 21 1 153 0 2 0 1 0.257 110 25 0 152 1 0
0 1 0.225 110 32 0 151 1 0 4 1 0.196 110 5 0 206 5 0 5 1 0.456 110 40 0 200 0 0 6 2 1.444 110 10 1 192 0 0 0 0 0.430 110 30
0 149 0 0 0 0 0.321 110 65 0 80 0 5 3 0 1.301 110 40 1 132 0 0 0 0 0.280 110 21 0 126 0 0 0 0 0.260 110 21 0 72 0 2 10 0
3.073 110 50 1 71 0 0 2 0 0.064 110 21 0 -1 18 4 5 2 1.031 110 90 1 70 1 0 2 0 0.026 110 100 0

(38) The “intensity of the exercise” data point indicates how intense the exercise is to the user, for example, intensity values may be: 1=normal (E.g. normal walk, etc.), 2=moderate (brisk walk, biking, etc.), and 3=intense (high intensity interval training (HIIT), running, etc.)

(39) In an example using the third row of Table 1 above, the exercise model may indicate that the physiological data in the third row triggers an unsafe exercise response as indicated by the label=1. The basis for the exercise model indicating that the physiological data indicates may be based for example on a low TDI of 21 Units (U), the 10 U insulin in past 30 minutes, and a current IOB of approximately 4.332 Units. Together, these parameters, when combined with the user's performance of intense exercise, or perhaps even mild exercise (given the particular user of a wearable drug delivery device), and when evaluated by the exercise model, may indicate that participation in exercise by the user is either safe or unsafe.

(40) The other rows of data that resulted in being labeled unsafe were based on various factors. For example, the row in which the CGM value is 200, the data that generated the unsafe to exercise indication was based on the low TDI (10), the 6 U of insulin in the past 30 mins, current IOB of approximately 1.444 U. Similarly, the third indication of unsafe to participate in exercise is in the CGM reading of 80, the data that generated the unsafe to exercise indication was based on a low CGM reading (80), a medium range TDI of approximately 40, the medium range IOB of approximately 1.301, and the 5 U requested bolus. The fourth indication of unsafe to participate in exercise is in the CGM reading of 72, the data that generated the unsafe to exercise indication was based on a low CGM reading (72), a medium range TDI of approximately 50, the higher IOB of approximately 3.703, the 10 U of insulin in the last 30 minutes, and the 2 U requested bolus. The fifth indication of unsafe to participate in exercise is in the CGM reading of -1, which suggests that CGM readings have been

missed past a threshold amount (for example, for the past 30 minutes), and in this case, for past 90 minutes (5 minutes times 18 missed readings), the medium range of IOB of approximately 1.031, the 5 U of insulin in past 30 mins, a requested bolus of 4 U and the TDI of 90.

(41) Based on the label, the programming code may cause the processor to generate an exercise safety signal indicating whether exercise is safe or unsafe. The processor may respond to the generated exercise safety signal as discussed above. The exercise model may be a machine learning model, such as a linear regression model, a logistic regression model, a decision tree model, a support vector machine, a naïve Bayes classifier, or the like.

(42) It may be helpful to describe at a high level, details related to the exercise model, which in the example below is a logistic regression model, and, in particular, how the exercise model may be built, trained, and modified, if necessary. The exercise model may be generated by a processor in a server, such as a cloud-based server, or the like.

(43) In an operational example of how the logistic regression exercise model is built; a processor may be further configured to obtain data from a dataset including physiological data indicating physiological conditions of the wearer. The processor may split data in the dataset into training data and validation data. Using the training data, a logistic regression model may be trained. The validation data may be used to validate an output of the logistic regression model. Based on a result of the validation of the output of the logistic regression model, the processor may perform hyperparameter tuning to improve the accuracy of the logistic regression model. The processor may retrieve the received data related to physiological conditions of the wearer and input the retrieved current data into the logistic regression model for evaluation. The processor may, for example, label the output data as either indicating an unsafe condition for exercise or a safe condition for exercise based on a value of output data from the logistic regression model. The processor may generate an exercise safety signal based on any label indicating an unsafe condition for exercise.

(44) Once you build a classifier and enable a Boolean output, the classifier may provide a probability related to whether it is safe to exercise. A threshold based on past history (whether it was safe to exercise in the past) may be established, and everything above threshold would be true, everything below would be false. Boolean value of true or 1 would be unsafe to exercise.

(45) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(46) FIG. 3 is a flowchart of a further example process implemented according to another aspect of the disclosed subject matter.

(47) A processor or logic circuitry may be operable to execute programming code stored in a memory coupled to the respective processor or logic circuitry. In block **302**, a processor executing process **300** may be operable to determine that a wearer of a wearable drug delivery device is participating in exercise. It is noted that exercise can have both short- and long-term physiological effects.

(48) Once participation in exercise is detected, the system can implement a long-term lengthening of the IOB curves over a period of 12-24 hours following exercise to account for the long-term effects of exercise. For example, as shown in block **304**, the process **300** modifies an insulin onboard curve in response to the determination that a wearer is participating in exercise. The processor when executing process **300** may modify the insulin onboard (IOB) curve by extending a length of the insulin onboard curve. Extending the length of the IOB curve slows decay of a calculated amount of insulin onboard and the extension remains in place for a preset period of time.

(49) The modification of the IOB curve makes the processor evaluate the effect of exercise based on changing the effective IOB (i.e., the amount of insulin onboard that will be metabolized by the wearer's body over time), while the wearer's sensitivity to insulin remains unchanged because a change to sensitivity would be a direct change to the exercise model. Instead, the modification of the length of the IOB curve makes a change to the perceived length of time the insulin is acting in the wearer's body. Based on the extended IOB curve, the processor may modify constraints that are applied to the insulin doses (e.g., limits on the maximum amount of insulin that is included in a bolus, delivered within X number of hours of the determined participation in exercise, or the like, where X is approximately 3.5 to approximately 4.5) but the medication delivery algorithm continues making insulin delivery decisions based on threshold values related to exercise safety and the modified constraints. For instance, the assumed 3-hour and 4-hour IOB decay curves may be lengthened to 3.5 hours and 4.5 hours following detection of significant exercise, to incorporate the impact of relatively longer-lasting increased insulin sensitivity. Of course, other values of X may be 1.5, 2, 6, 12 or the like. This is illustrated in the example IOB curves of FIG. 5.

(50) In block **306**, the processor implementing process **300** may use the modified insulin onboard curve to calculate a next dosage of insulin to be delivered to the wearer and calculate a time to deliver the next dosage.

(51) In block **308**, the process **300** may cause the next dosage of insulin to be delivered at the calculated time to the wearer.

(52) Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

(53) FIG. 4 illustrates a process **400** in accordance with an embodiment of the disclosed subject matter.

(54) In block **402**, a processor executing process **400** may be operable to, in response to the determination that the wearer of a wearable drug delivery device is participating in exercise, determine a current insulin onboard of the wearer.

(55) In block **404**, process **400** compares the determined current insulin onboard (IOB) to a threshold value set for the wearer. The threshold value for insulin onboard may be one of the threshold values related to exercise safety that is evaluated by the processor. In this example, the current IOB may be a primary factor used in the determination of whether participation in exercise is safe or unsafe for the wearer of the wearable drug delivery device. For example, an IOB of 3 U or greater may indicate that the artificial pancreas application is overcompensating or overdosing insulin, and hence it is

unsafe for the user to exercise. Further, the processor may also determine whether the wearer's total daily insulin (TDI) is going to reach a threshold TDI value, such as, for example, 4 times, 5 times, or 9 times the expected total daily insulin delivery in a 3-hour period of time or another period of time, such as 2-hour period of time, and hence make a determination that it is unsafe to exercise. Alternatively, another threshold may be one in which the total insulin delivered over the last 3 hours, including manual boluses and system automated deliveries, is not permitted to exceed 15 times the number of basal hours, or 31% of TDI. Of course, other TDI thresholds that indicate overcompensation or over delivery may be used.

(56) In block **406**, process **400** in response to the current insulin onboard being above the threshold value set for the wearer, generates an exercise safety signal indicating an unsafe exercise condition. In block **408**, process **400** based on the generated exercise safety signal, suspends delivery of insulin for a period of time and/or reducing all manual bolus dosage amounts by a preset percentage, such as 25%, 50%, 75%, or in some cases 100%.

(57) FIG. 5 is a modified insulin onboard decay curve that illustrates an example usable in an embodiment of the disclosed subject matter.

(58) In modified insulin onboard decay curve **500** of FIG. 5, the solid line represents the original 3-hour curve, the dashed line represents the 3-hour insulin decay curve that has been extended by 30 minutes (i.e., the 3.5-hour insulin delay curve), the 4-hour insulin delay curve is shown as dash-dot-dash, and the dashed-dot-dot-dashed line represents the respective 4-hour insulin decay curve extended by 30 minutes (i.e., the 4.5-hour insulin delay curve). As can be seen in the curves, for example, at the 100.sup.th minute mark, the remaining insulin action is approximately 36% with the 3-hour insulin delay curve, but approximately 46% with the 3.5-hour insulin delay curve, resulting in increased estimated remaining insulin on board and thus potentially a more conservative insulin delivery by the algorithm. If IOB is higher, the insulin delivery algorithm will tend to deliver less insulin.

(59) The modifications of the IOB curves may vary based on the system's original IOB curves that are utilized. The original IOB curves may be based on anonymized, clinical data gathered from hundreds or thousands of patients or based on a specific wearer's history.

(60) FIG. 6 illustrates an example of a process for generating an exercise model usable with the aspects of the disclosed subject matter. The generation of the exercise model may occur in a server, such as a server provided by cloud-based services or the like, that is also communicatively coupled to the controller as described with reference to a later example.

(61) As previously mentioned, the exercise model may be a machine learning model that is capable of modifying its output based on the values input into the model and a closeness of the output value to a result produced from a validation data set. In the following discussion of routine **600**, the exercise model is implemented as a logistic regression model.

(62) In block **602**, a server executing routine **600** may obtain data from a dataset that includes physiological data indicating physiological conditions of the wearer. For example, the controller may be operable to upload anonymized physiological data indicating the physiological conditions of the wearer to the cloud-based services. The data uploaded from the controller may be combined with similar anonymized data that is uploaded from other controllers. Alternatively, the controller may be operable to obtain and process the data. For example, the controller may obtain the data from a memory coupled to the controller or from a server or data storage accessible through cloud-based services. The obtained data may be an entire data set or part of a data set, such as a past evaluation period which may be three, six, nine days, or the like.

(63) In block **604**, the server or the controller may split data in the dataset into training data and validation data.

(64) In block **606**, a logistic regression model may be trained by the server or the controller using the training data.

(65) In block **608**, the server or the controller may use the validation data to validate an output of the logistic regression model.

(66) Based on a result of the validation of the output of the logistic regression model, the controller or server may in block **610** perform hyperparameter tuning to improve the accuracy of the logistic regression model. Once the exercise model is appropriately trained, the controller may begin executing the exercise model to determine whether it is safe or unsafe for the wearer to participate in exercise based on the wearer's physiological conditions.

(67) For example, the controller may in block **612** be operable to utilize the exercise model in the determination of whether it is safe or unsafe for a wearer of a wearable drug delivery device to participate in exercise.

(68) In the examples described with reference to FIGS. 1-6, a controller may input data related to the physiological conditions of the wearer into the exercise model and obtain an exercise safety signal. The data related to the physiological condition of the wearer may be blood glucose measurement values, IOB, total daily insulin (TDI), and similar data as described above with reference to other examples.

(69) FIG. 7 illustrates a functional block diagram of a system example suitable for implementing the example processes and techniques described herein.

(70) The automatic wearable drug delivery system **700** may implement (and/or provide functionality for) a medication delivery algorithm (MDA), such as an artificial pancreas (AP) application, to govern or control automated delivery of a drug, a therapeutic, or a medication, such as insulin, to a user (e.g., to maintain euglycemia—a normal level of glucose in the blood). The automatic wearable drug delivery system **700** may, for example, include an analyte sensor **704**, a controller **706**, a wearable drug delivery device **708**, and an optional smart accessory device **702**.

(71) The controller **706** may be remote from the wearable drug delivery device **708** and may include a user interface **716**, a communication device **722**, a memory **712**, and a processor **714**. The user interface **716** is coupled to the processor **714** and operable to receive inputs related to a physiological condition of a wearer and provide the input to the processor **714**. In an example, the input may be a request for a bolus dosage. The controller **706** may include a user interface **716**, which may be a keypad, a touchscreen display, levers, light-emitting diodes, buttons on the controller **706**, a microphone, a camera, a speaker, a display, or the like, that is configured to allow a user to enter information and allow the controller **706** to output

information for presentation to the user (e.g., alarm signals, exercise recommendations (e.g., exercise times and/or exercise intensity, and the like). The user interface **716** may provide inputs, such as a voice input, a gesture (e.g., hand or facial) input to a camera, swipes to a touchscreen, or the like, to processor **714** which the programming code interprets.

(72) The controller **706** may be a computing device such as a smart phone, a tablet, a personal diabetes controller, a dedicated diabetes therapy controller, or the like. In an example, the controller **706** may include a processor **714**, a controller memory **712**, a user interface **716**, and a communication device **722**. The controller **706** may contain analog and/or digital circuitry that may be implemented as a processor **714** for executing processes based on programming code stored in the controller memory **712**, such as the medication delivery algorithm or application (MDA) **710** and/or the exercise model **782** and related programming code as well as threshold values related to exercise safety. In addition, the memory **712** may store programming code to manage a user's blood glucose levels and for controlling the delivery of the drug, medication, or therapeutic agent to the user as well as programming code that determines via the exercise model **782** whether it is safe or unsafe for a wearer (or user) to participate in exercise and cause the execution of different actions in response to the determination, as discussed above. The controller **706** may be used to program, adjust settings, and/or control operation of the wearable drug delivery device **708** and/or the analyte sensor **704** as well as the optional smart accessory device **702**.

(73) The one or more transceivers, transceiver A **718** and transceiver B **720** may operate according to one or more radio-frequency protocols. In the example, the transceivers **718** and **720** may be a cellular transceiver and a Bluetooth® transceiver, respectively. For example, the transceiver A **718** or transceiver B **720** may be configured to receive and transmit signals containing information usable in the exercise model **782** by the MDA **710**, such as physiological conditions of a wearer, other physiological data, IOB curves and the like.

(74) The wearable drug delivery device **708** may include logic circuitry **748**, a reservoir **738**, a communication device **742**, a power source **740**, a memory **712**, user interface (UI) **750**, and a pump mechanism **744**. The logic circuitry **748** may be operable to control the drug delivery device. The reservoir **738** may be configured to contain a liquid drug. The communication device **742** may be coupled to the logic circuitry **748**. The pump mechanism **744** may be responsive to the logic circuitry **748** and fluidically coupled to the reservoir **738**.

(75) The memory **712** may store programming code executable by the processor **714**. The programming code, for example, may enable the processor **714** to control expelling insulin from the reservoir **738** in response to control signals from the controller **706** and MDA **710** or based on signals from the optional MDA **780**.

(76) In the example, the communication device **742**, which may be a receiver, a transmitter, or a transceiver that operates according to one or more radio-frequency protocols, such as Bluetooth, Wi-Fi, a near-field communication standard, a cellular standard, or the like. The communication device **764** may enable the logic circuitry **748** to communicate with the controller **706** and the analyte sensor **704**.

(77) The wearable drug delivery device **708** may be attached to the body of a user, such as a patient or diabetic, at an attachment location and may deliver any therapeutic substance to a user at or around the attachment location. For example, a bottom surface of the wearable drug delivery device **708** may include an adhesive to facilitate attachment to the skin of a wearer.

(78) The reservoir **738** may store drugs, medications or therapeutic agents suitable for automated delivery, such as diabetes treatment drugs (e.g., insulin, glucagon, glucagon-like peptides), pain relief drugs (e.g., morphine), hormones, blood pressure medicines, chemotherapy drugs, or the like. The wearable drug delivery device **708** may include a needle or cannula (not shown in this example) coupled to the reservoir **738** and extending into the body of the wearer for delivering the drug into the wearer's body of the user (which may be done subcutaneously, intraperitoneally, or intravenously), and a pump mechanism **744** under control of the logic circuitry **748** transfers the drug from the reservoir **738** through the needle or cannula and into the wearer.

(79) The power source **740**, such as a battery, a piezoelectric device, other forms of energy harvesting devices, or the like, for supplying electrical power to the pump mechanism **744** and/or other components of the wearable drug delivery device **708**.

(80) In some examples, the wearable drug delivery device **708** may include a user interface **750**, which may be a keypad, a touchscreen display, levers, light-emitting diodes, buttons on a housing of the drug delivery device **708**, a microphone, a camera, a speaker, a display, or the like, that is configured to allow a user to enter information and allow the drug delivery device **708** to output information for presentation to the user (e.g., audio, visual or vibrational alarm signals or the like). The user interface **750** may provide inputs, such as a voice input, a gesture (e.g., hand or facial) input to an optical sensor, swipes to a touchscreen, or the like, to processor **714** which the programming code interprets.

(81) The wearable drug delivery device **708** may optionally include a device sensor **754** that may include an accelerometer, a gyroscope, a skin conductance measuring device (e.g., to measure perspiration due to exercise), or the like. Signals from the device sensor **754** may be provided to the controller **706** for use in determining whether a wearer is participating in exercise.

(82) The smart accessory device **702** may be a smart watch, another wearable smart device, including eyeglasses, provided by other manufacturers, a global positioning system-enabled wearable device, a wearable fitness device, smart clothing, or the like, and may be operable to communicate with the other components of system **700** via wireless communication links **774**, **776**, or **778**.

(83) For example, the smart accessory device **702** may include a communication device **736**, a processor **734**, a user interface **732** and a memory **730**. The user interface **732** may be a graphical user interface presented on a touchscreen display of the smart accessory device **702**. The memory **730** may store programming code to operate different functions of the smart accessory device **702** as well as an instance of the MDA **728**. The processor **734** that may execute programming code, such as MDA **728** for controlling the wearable drug delivery device **708** to implement the processes and techniques of

FIGS. 1-6 described herein.

(84) The analyte sensor **704** may include a processor **758**, a memory **762**, a sensing/measuring device **766**, a user interface **756**, a power source **752**, and a communication device **766**. The analyte sensor **704** may, for example, be a blood glucose monitor removably attachable via adhesive, for example, to a body of the wearer. In such an example, the analyte sensor **704** is operable to measure a blood glucose measurement value of the wearer (not shown) and communicate with the controller **706** and the drug delivery device **708** via the communication device **764** under the control of the processor **758**. The memory **730** may be configured to store information and programming code, such as an instance of the MDA **760**.

(85) The analyte sensor **704** may be configured to detect one or more different analytes, such as lactate, ketones, uric acid, sodium, potassium, alcohol levels, blood glucose, proteins, hormones, or the like, and output results of the detections, such as measurement values or the like, for receipt by one or more of **702**, **706** or **708**. The analyte sensor **704** may, in an example, be configured to measure the blood glucose value at a predetermined time interval, such as every 5 minutes, or the like. The communication device **764** of analyte sensor **704** may have circuitry that operates as a transceiver for communicating the measured blood glucose values to the controller **706** over a wireless link **768** or with the wearable drug delivery device **708** over the wireless communication link **772**. While called an analyte sensor **704**, the sensing/measuring device **533** of the analyte sensor **704** may include one or more additional sensing elements, such as a heart rate monitor, a pressure sensor, or the like.

(86) The processor **758** of the analyte sensor **704** may be operable to perform many functions. For example, the programming code stored in the memory **762** may enable the processor **758** to manage the collection and analysis of data detected by the sensing and measuring device **766**, such as blood glucose measurement values, providing trend information and the like. The processor **758** may include discrete, specialized logic and/or components, an application-specific integrated circuit, a microcontroller or processor that executes software instructions, firmware, programming instructions stored in memory (such as memory **762**), or any combination thereof.

(87) Services provided by cloud-based services **724** may include data storage that stores anonymized data, such as blood glucose measurement values, historical IOB or TDI, maximum and minimum boundary values, therapeutic exogenous substance concentration values, substance sensitivity values, and other forms of data. In addition, the cloud-based services **724** may process the anonymized data from multiple users to provide generalized information related to TDI, insulin sensitivity, JOB, and the like. The cloud-based services **724** may be accessed via data network device **786**, which may be a Wi-Fi device, a cellular communication tower, a local area network, a campus wide network or the like.

(88) The wireless communication links **726**, **768**, **770**, **772**, **774**, and **776** may be any type of wireless link operating using known wireless communication standards or proprietary standards. As an example, the wireless communication links **726**, **768**, **770**, **772**, **774**, and **776** may provide communication links based on Bluetooth®, Zigbee®, Wi-Fi, a near-field communication standard, a cellular standard, or any other wireless protocol via the respective communication devices communication links **726**, **768**, **770**, **772**, **774**, and **776**.

(89) Software related implementations of the techniques described herein, such as the processes examples described with reference to FIGS. 1-6 may include, but are not limited to, firmware, application specific software, or any other type of computer readable instructions that may be executed by one or more processors. The computer readable instructions may be provided via non-transitory computer-readable media. Hardware related implementations of the techniques described herein may include, but are not limited to, integrated circuits (ICs), application specific ICs (ASICs), field programmable arrays (FPGAs), and/or programmable logic devices (PLDs). In some examples, the techniques described herein, and/or any system or constituent component described herein may be implemented with a processor executing computer readable instructions stored on one or more memory components.

(90) In addition, or alternatively, while the examples may have been described with reference to a closed loop algorithmic implementation, variations of the disclosed examples may be implemented to enable open loop use. The open loop implementations allow for use of different modalities of delivery of insulin such as smart pen, syringe or the like. For example, the disclosed AP application and algorithms may be operable to perform various functions related to open loop operations, such as the generation of prompts requesting the input of information such as weight or age. Similarly, a dosage amount of insulin may be received by the AP application or algorithm from a user via a user interface. Other open-loop actions may also be implemented by adjusting user settings or the like in an AP application or algorithm.

(91) Some examples of the disclosed device or processes may be implemented, for example, using a storage medium, a computer-readable medium, or an article of manufacture which may store an instruction or a set of instructions that, if executed by a machine (i.e., processor or controller), may cause the machine to perform a method and/or operation in accordance with examples of the disclosure. Such a machine may include, for example, any suitable processing platform, computing platform, computing device, processing device, computing system, processing system, computer, processor, or the like, and may be implemented using any suitable combination of hardware and/or software. The computer-readable medium or article may include, for example, any suitable type of memory unit, memory, memory article, memory medium, storage device, storage article, storage medium and/or storage unit, for example, memory (including non-transitory memory), removable or non-removable media, erasable or non-erasable media, writeable or re-writeable media, digital or analog media, hard disk, floppy disk, Compact Disk Read Only Memory (CD-ROM), Compact Disk Recordable (CD-R), Compact Disk Rewritable (CD-RW), optical disk, magnetic media, magneto-optical media, removable memory cards or disks, various types of Digital Versatile Disk (DVD), a tape, a cassette, or the like. The instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, encrypted code, programming code, and the like, implemented using any suitable high-level, low-level, object-oriented, visual, compiled and/or interpreted programming language. The non-transitory computer readable medium embodied programming

code may cause a processor when executing the programming code to perform functions, such as those described herein.

(92) Certain examples of the present disclosure were described above. It is, however, expressly noted that the present disclosure is not limited to those examples, but rather the intention is that additions and modifications to what was expressly described herein are also included within the scope of the disclosed examples. Moreover, it is to be understood that the features of the various examples described herein were not mutually exclusive and may exist in various combinations and permutations, even if such combinations or permutations were not made express herein, without departing from the spirit and scope of the disclosed examples. In fact, variations, modifications, and other implementations of what was described herein will occur to those of ordinary skill in the art without departing from the spirit and the scope of the disclosed examples. As such, the disclosed examples are not to be defined only by the preceding illustrative description.

(93) Program aspects of the technology may be thought of as “products” or “articles of manufacture” typically in the form of executable code and/or associated data that is carried on or embodied in a type of non-transitory, machine readable medium. Storage type media include any or all the tangible memory of the computers, processors or the like, or associated modules thereof, such as various semiconductor memories, tape drives, disk drives and the like, which may provide non-transitory storage at any time for the software programming. It is emphasized that the Abstract of the Disclosure is provided to allow a reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features are grouped together in a single example for streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate example. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein,” respectively. Moreover, the terms “first,” “second,” “third,” and so forth, are used merely as labels and are not intended to impose numerical requirements on their objects.

(94) The foregoing description of examples has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present disclosure to the precise forms disclosed. Many modifications and variations are possible considering this disclosure. It is intended that the scope of the present disclosure be limited not by this detailed description, but rather by the claims appended hereto. Future filed applications claiming priority to this application may claim the disclosed subject matter in a different manner and may generally include any set of one or more limitations as variously disclosed or otherwise demonstrated herein.

Claims

1. A wearable drug delivery device, comprising: a processor, a reservoir configured to contain insulin; a pump mechanism communicatively coupled to the processor and fluidically coupled to the reservoir; and a memory coupled to the processor and operable to store programming code that, when executed by the processor, causes the processor to: use an insulin on board (IOB) curve in determining a basal dose of insulin to deliver to a wearer; receive data indicative of a physiological condition of the wearer of the wearable drug delivery device; evaluate the received data with relation to the wearer participating in exercise; based on a result of the evaluation, determine whether it is safe for the wearer to participate in exercise; and output an exercise safety signal indicating a result of the determination of whether it is safe or unsafe for the wearer to participate in exercise; and where the evaluation determines that the wearer is exercising, lengthening the IOB curve, and using the lengthened IOB curve to determine a next basal dose of insulin for the wearer.
2. The wearable drug delivery device of claim 1, wherein the processor, when determining whether it is safe for the wearer to participate in exercise, is further operable to: determine whether the result of the evaluation indicates a presence of one or more conditions; and based on the determination of the presence of the one or more conditions, generate an indication that exercise is unsafe.
3. The wearable drug delivery device of claim 2, wherein the one or more conditions include a blood glucose measurement value, a total daily insulin amount, a request for a bolus or a request to use a hypoglycemia protect mode, a current calculation of an amount of insulin onboard, or an amount of insulin delivered in a past evaluation period.
4. The wearable drug delivery device of claim 1, wherein the processor, when determining whether it is safe for the wearer to participate in exercise, is further operable to: determine whether the result of the evaluation indicates that a trajectory of blood glucose measurement values of the wearer is trending downward; and based on the determination that the trajectory of the glucose measurement values of the wearer are trending downward, generate an indication that exercise is unsafe.
5. The wearable drug delivery device of claim 1, wherein the processor, when executing the stored programming code, is further operable to: retrieve physiological data from the memory, wherein the physiological data contains a number of days of data related to the wearer's participation in exercise; and update an exercise model using the retrieved physiological data.
6. The wearable drug delivery device of claim 5, wherein the processor, when evaluating the received data related to the wearer participating in exercise, is further operable to: input values obtained from the received data in the exercise model; and analyze an output of the exercise model with reference to threshold values related to exercise safety for the wearer.
7. The wearable drug delivery device of claim 1, wherein the received data includes a current blood glucose measurement value, a total daily insulin value, a time and amount of a requested bolus, a current insulin onboard value, or an amount of insulin delivered within a past evaluation period.
8. The wearable drug delivery device of claim 7, wherein the past evaluation period is a 15 minute period, a 30 minute

period, a 60 minute period or a 120 minute period of time.

9. The wearable drug delivery device of claim 1, further comprising: a communication device, wherein the communication device is operable to provide a wireless communication connection with an external device; and the processor is further operable to: control the communication device; establish the wireless communication connection with the external device, wherein the external device is a data network device; retrieve physiological data from a memory source, wherein the physiological data contains a number of days of data related to the wearer's participation in exercise; and update an exercise model using the retrieved physiological data.

10. The wearable drug delivery device of claim 1, wherein the processor is further configured to: obtain data from a dataset including physiological data related to physiological conditions of the wearer; split data in the dataset into training data and validation data; train a logistic regression model using the training data; use the validation data to validate an output of the logistic regression model; based on a result of the validation of the output of the logistic regression model, perform hyperparameter tuning to improve accuracy of the logistic regression model; retrieve the obtained data related to the physiological conditions of the wearer; input the retrieved data into the logistic regression model for evaluation; based on a value of output data from the logistic regression model, label the output data as either indicating an unsafe condition for exercise or a safe condition for exercise; and generate the exercise safety signal based on any label indicating the unsafe condition for exercise.

11. The wearable drug delivery device of claim 1, further comprising: an output device coupled to the processor, wherein the processor is further operable to: interpret the outputted exercise safety signal; and provide an output signal to the output device based on the outputted exercise safety signal.

12. The wearable drug delivery device of claim 1, wherein the processor is further operable to: modify a medication treatment plan based on the outputted exercise safety signal.

13. A wearable drug delivery system, comprising: a drug delivery device, including: logic circuitry operable to control the drug delivery device; a reservoir configured to contain insulin; a communication device coupled to the logic circuitry; and a pump mechanism responsive to the logic circuitry and fluidically coupled to the reservoir; and a controller, including: a processor; a memory coupled to the processor and configured to store a medication delivery application, an exercise safety application, and programming code that, when the exercise safety application is executed by the processor, the processor is operable to: receive an input related to a physiological condition of a wearer, wherein the input is from the wearer; evaluate data parameters related to physiological conditions of a wearer and the received input related to the physiological condition of a wearer; determine whether the wearer has participated in exercise; in response to a determination that the wearer has participated in exercise and a result of the evaluation of the data parameters, generate an exercise safety signal; present an indication of a result of an interpretation of the exercise safety signal, wherein the indication is a safe or unsafe indicator; modify a medication treatment plan in response to the generation of the exercise safety signal, wherein the modifying includes lengthening an insulin on board (IOB) curve and using the lengthened IOB curve to determine basal insulin doses for a period of multiple hours.

14. The wearable drug delivery system of claim 13, wherein the data parameters related to physiological conditions of a wearer include one or more of: a blood glucose measurement value, a total daily insulin amount, a request for a bolus, a request to use a hypoglycemia protect mode, a current calculation of an amount of insulin onboard, or an amount of insulin delivered in a past evaluation period.

15. The wearable drug delivery system of claim 13, wherein the controller is a device separate from the drug delivery device.

16. The wearable drug delivery system of claim 13, wherein the controller further comprises: a user interface communicatively coupled to the processor and operable to receive the input related to a physiological condition of a wearer and provide the input to the processor, wherein the input is a request for a bolus dosage.

17. The wearable drug delivery system of claim 13, further comprising: an analyte sensor configured to be attached to a body of the wearer, and including an analyte processor and an analyte communication device, wherein the analyte sensor is operable to: measure a blood glucose measurement value of the wearer, and communicate with the controller and the drug delivery device via the analyte communication device under control of the analyte processor.

18. The wearable drug delivery system of claim 13, wherein the processor, when evaluating the data parameters, is further operable to: input data parameters including a blood glucose measurement value, a total daily insulin value, a time and an amount of a requested bolus, an amount of insulin onboard value, or an amount of insulin delivered within a past evaluation period into an exercise model; and based on a result of an output from the exercise model, generate the exercise safety signal.

19. A non-transitory computer-readable storage medium, the computer-readable storage medium including instructions that when executed by a processor, cause the processor to: determine that wearer of a wearable drug delivery device is participating in exercise; lengthen an insulin on board curve in response to the determination that a wearer is participating in exercise; use the lengthened insulin on board curve to calculate subsequent doses of insulin to be delivered to the wearer during a period of multiple hours.

20. The non-transitory computer-readable storage medium of claim 19, further including instructions that when executed by the processor, cause the processor to: in response to the determination that the wearer of a wearable drug delivery device is participating in exercise, determine a current insulin onboard of the wearer; compare the determined current insulin onboard to a threshold value set for the wearer; in response to the current insulin onboard being above the threshold value set for the wearer, generate an exercise safety signal indicating an unsafe exercise condition; and based on the generated exercise safety signal, suspend delivery of insulin for a period of time.

